

Highlights

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

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- Scoring the same model at the fire, not region-wide, costs 0.143 AUC
- The cause is the negative pool, not prevalence: +0.155 against -0.000
- Under blocked CV, thermal predictors add +0.06 to +0.15 AUC in five regions
- Equalising the evaluation frame lifts mean transfer from 0.540 to 0.617
- None of 20 transferability diagnostics is shown to order transfer skill

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

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Abstract

Pre-fire thermal dryness separates a fire year from a normal year, but models built on it are rarely tested outside the region where they were fitted. Five Mediterranean wildfire regions were analysed on about 500 m cells, with MCD64A1 labels and spatially blocked validation, adding six thermal predictors to a terrain, fuel and greenness baseline. The first result is about **evaluation geometry**: holding the model, its predictors and its fitting fixed and changing only which cells are scored, moving from the whole region to the burn scar and its 2 km collar costs **0.143 ROC-AUC** [**+0.077, +0.208**], which is the size of the predictor-block increments this literature reports as findings. A control isolates the cause as the composition of the negative pool rather than class balance: matching prevalence costs -0.000 [$-0.003, +0.002$] while substituting fire-adjacent negatives costs $+0.155$ [$+0.093, +0.217$] over nine held-out scars. Applied between regions the same effect withdrew five of our own claims. The five study areas enclose very unequal far fields, and restricting each to a 10 km collar, which removes only far-field negatives, lifts mean transfer from 0.540 to 0.617 and leaves no sign reversal supported under

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our own criterion, though the LST anomaly still differs between regions on a difference interval. The agreed direction is the opposite of the one dryness physics predicts: at the point estimates hotter pre-fire surfaces burned less in every region, and on mutual adjustment temperature survives where greenness does not, so the absolute channels behave here as static land-surface descriptors. Local skill nonetheless does not travel. The thermal block adds +0.056 to +0.153 within regions at 1 km blocking, +0.045 to +0.148 at 5 km, but +0.022 $[-0.032, +0.077]$ when a whole burn scar is withheld and +0.004 $[-0.028, +0.036]$ across twenty transfer directions, with a sign that varies by pair; the static baseline transfers no better, at 0.537 against 0.541 as drawn and 0.593 against 0.617 equalised; and equalised transfer of 0.617 sits against a matched within-region reference of 0.772, a shortfall of 0.155. Of twenty candidate diagnostics from five families, none was shown to order transfer: eighteen failed on the frames as drawn, and the two that succeeded require target labels and stop ordering transfer once the frames are equalised, one losing its variance entirely and the other its correlation (ρ +0.81 to -0.06). Transfer skill has to be measured, not inferred from similarity.

Keywords: wildfire, model transferability, area of applicability, land surface temperature, domain adaptation, concept shift

1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns [29]. The pressure to anticipate *where* fire will occur has produced a large and still-growing literature on fire susceptibility mapping. It has recently been synthesised from

both a methodological and an application perspective [42, 14]. The dominant pattern is well established. Geospatial predictors are assembled over a study region and paired with a historical record of burned area. A supervised classifier is then fitted, most often a random forest [5]. The resulting
10 susceptibility surface is published with a cross-validated AUC in the 0.85 to 0.95 range.

This paper is about a property that the pattern does not report. Predictors that describe the *state* of a surface in a particular season are more informative about that season than predictors that describe a place. They
15 are also, as we show, not portable, and a large part of this paper is about how much harder that is to establish than it looks: the reading we first reached, that the relationship is reparameterised locally and points in different directions in different places, does not survive a correction to how the study regions were evaluated (Section 4.4). The dynamic block is worth +0.056 to
20 +0.153 ROC-AUC inside every region under blocked cross-validation, +0.022 when a whole burn scar is withheld, and contributes +0.004 between them, an estimate whose interval spans zero, with a sign that is a property of the source-target pair rather than of the block.

The obvious reading of that pair of numbers is a trade-off, and our own
25 earlier framing took it. The data do not support it. The static baseline transfers at a mean of 0.537 against the thermal model’s 0.541, so the predictor class that ought to travel does not travel either, and removing the reversing predictors returns +0.014 of transfer on an interval that also spans zero. Two
nulls on the portability axis are not an exchange. What is measured is a local
30 gain that does not export, and a local cost when the reversing predictors are

removed, with no compensating transfer gain on either arm.

1.1. Portability goes unmeasured

A susceptibility model’s reported skill is almost always an estimate of *within-region* performance. Held-out folds come from the same study area
35 and season, often from the same fire event, and where folds are drawn at random over cells, spatial autocorrelation inflates the estimate further. The problem is documented across ecological modelling [33, 31] and is addressed by spatially blocked cross-validation [39, 25]. Blocking is the right correction for what it corrects, but it does not make the estimate honest about a fire
40 the model has not seen, and Section 4.3 measures that gap on identical cells.

Because only that side of the ledger is reported, portability is never entered at all. A predictor block is adopted on the strength of the increment it delivers inside its training footprint, and whether that increment survives a change of region is not asked, even though any regional product built from
45 locally trained models implicitly promises generalisation beyond it. For fire the transfer question is asked far less often than in adjacent fields. Meteorological fire-danger indices are known not to port cleanly between fire environments [32], and the two recent studies that test model transfer systematically, across Mediterranean countries [8] and US counties [16], both
50 report that it largely succeeds between similar regions — and both transfer models whose dominant predictors are *spatially stationary*. Whether a dynamic, season-specific predictor class behaves the same way is the question this paper puts.

1.2. Pre-fire thermal dryness is the natural test case

55 The dynamic class used here is pre-fire thermal dryness, and it is the class most plausibly *expected* to transfer. Standard susceptibility predictors are static or near-static over the timescale at which fire danger varies, and such a set explains poorly why one summer burned and the preceding one did not. What changes between those summers is the state of the surface, 60 to which satellite thermal observation gives partial access (Section 2.2). The Temperature-Vegetation Dryness Index [35] formalises that access into an internally normalised measure which should, in principle, be less exposed to absolute-temperature offsets between regions than raw land surface temperature. **That theoretical portability advantage is tested here and is** 65 **not found:** the two internally normalised channels transfer no better than the four absolute ones (Appendix A(f)).

The physics linking moisture stress to combustion is universal, so portability should be most expected for this class, which is what makes it diagnostic: a loss here cannot be dismissed as a peculiarity of a locally defined covariate. That expectation is the motivation for the design, not a finding of it, 70 and Section 4.4 reports that the signed associations run the other way — a hotter pre-fire surface is associated with *less* burning in all five regions, and on mutual adjustment temperature survives where greenness does not, so on this cohort the absolute channels behave as static land-surface descriptors 75 rather than as a dryness index. We keep the framing because it is why these predictors were chosen and why their failure to travel is informative, and we state the contradiction where the evidence appears rather than adjusting the motivation after the fact.

1.3. Why the loss is invisible to the diagnostics in use

80 A deficit on transfer can arise two ways. Under **covariate shift** the marginal distribution of the predictors differs while the predictor-response relationship is preserved, which is in principle correctable without target labels, by per-region standardisation or covariance alignment such as CORAL [37], an approach with an established remote-sensing literature [38]. Under
85 **concept shift** the conditional relationship itself changes: a predictor positively associated with burning in one region is negatively associated with it in another. No label-free realignment of input distributions can repair a sign reversal, because the information needed to detect it is exactly the information being withheld. The taxonomy is canonical [27].

90 This distinction is what makes the failure hard to anticipate. The diagnostics the field relies on to decide whether a model may be applied to new ground, above all area-of-applicability measures built on dissimilarity in predictor space [23, 24, 18], are marginal by construction. A target region can sit well inside the training data’s predictor envelope while the relation-
95 ship between those predictors and burning points the other way. This paper measures both sides on the same pairs.

1.4. Contributions

Three findings carry this paper, stated here as claims and established in Section 4, which carries every interval.

100 **Contribution 1. Where a model is scored decides what it appears to know, and the effect is large enough to dissolve findings of our own.** That evaluation extent inflates AUC is established in species distribution modelling (Section 2.5); what is new is a magnitude on the wildfire

problem under a controlled design. Holding the model, the predictors and
 105 the fitting fixed and changing only which cells are scored costs **0.143 ROC-AUC**, the size of the predictor-block increments this literature publishes as findings, with the cause isolated to the composition of the negative pool rather than class balance (Section 4.3). Applied between regions the same effect withdraws five claims of our own, including the sign reversal we had
 110 offered as the mechanism of the transfer residual and the one arm that held place fixed (Section 4.4). The practical consequence is a reporting standard (Section 5.8).

Contribution 2. Local skill does not travel, and correcting the frame does not rescue it. The thermal block is worth a substantial within-
 115 region increment in every one of five regions under blocked cross-validation, with every bootstrap interval above zero, and it remains positive in all five when the frame is equalised. Much of that is a property of interleaved hold-out: withholding a whole burn scar leaves an increment whose interval spans zero. Across twenty ordered transfer directions its paired contribution also
 120 spans zero, with a sign that is a property of the pair rather than of the block. Two controls bound the reading. The static baseline transfers no better than the dynamic one on either frame, so the failure is not the thermal block’s peculiarity; and on matched frames and matched blocking the equalised transfer still falls **0.155** short of the within-region reference. The
 125 within-region half is not novel [1, 13]; the paired contrast against portability is. Dimarco et al. [8] transfer a predominantly stationary predictor set successfully across a comparable Mediterranean design, though their response variable is ignition rather than burned area (Section 5.7).

Contribution 3. The shortfall cannot be anticipated by any
130 **diagnostic we could run.** Twenty candidate diagnostics from five families are evaluated against observed transfer under one bootstrap framework, including predictor-space dissimilarity [23, 24, 18], climatic and geographic distance, learned domain separability, niche overlap and burn-pattern regime distances. Eighteen were not shown to order the matrix on the frames as
135 drawn. Two conditional variants did, but both need burned labels on both sides, both rest on a data-selected feature subset whose all-nine counterparts span zero, and **Contribution 1 removes even those:** on comparable frames one becomes unanimous with no variance left to correlate and the other loses its correlation entirely (Section 4.4). The point survives without
140 any ranking: the pair with the highest burned-niche overlap is among the weakest in the matrix while the lowest transfers above chance in both directions, contradicting an expectation the fire literature carries [8, 16] and agreeing with species distribution modelling [41, 34].

Two consequences follow, reported as supporting results. Label-free align-
145 ment by standardisation and covariance alignment [37] does not repair transfer, and we believe this is the first application of covariance alignment to fire susceptibility. And because the residual is conditional, the resource that closes it is target labels, whose price is measured in Section 4.12.

Alongside these, a leakage-audited, spatially blocked evaluation and trans-
150 fer protocol is released with code, configuration and frozen outputs, so that most of this result can be re-run rather than taken on trust. One component is an exception and is named in the data-and-code availability statement: the reproduction-check code is not in the released repository at the commit

of record. The few-shot export, previously also listed as an exception, was
155 regenerated at commit `6d7a6a71` and its validator now passes on every check.
This matters given evidence that wildfire transfer conclusions are sensitive
to evaluation design and task formulation [45].

A companion paper treats the observational layer beneath this one, where
preprocessing decisions taken before any model is fitted are shown to carry
160 their own uncertainty budget.

Section 2 reviews the relevant literature. Section 3 describes the study
regions, the data, and the modelling, cross-validation, transfer and adapta-
tion protocol. Section 4 reports results. Section 5 discusses them, including
limitations. Section 6 concludes.

165 **2. Related work**

2.1. Fire susceptibility mapping, and what it reports

Fire susceptibility mapping has been synthesised recently from both method-
ological and application perspectives [42, 14]. The dominant pattern is sta-
ble: geospatial predictors are assembled over a study region, paired with a
170 historical burned-area record, and fitted with a supervised classifier, most
often a random forest [5], and the resulting surface is published with a cross-
validated AUC in the 0.85 to 0.95 range. That figure is a within-region
estimate. Comparable within-region results already exist for the landscapes
studied here [1, 13].

175 *2.2. Pre-fire thermal dryness*

Satellite thermal observation gives repeated access to surface state through
fuel moisture content [47]. Pairing land surface temperature with a vegeta-

tion index was established as a live-fuel-moisture estimator for fire-danger rating by Chuvieco et al. [7], and the Temperature-Vegetation Dryness Index [35] formalises that feature space into an internally normalised measure, which is why it is often expected to travel between regions better than raw temperature. That pre-fire thermal state carries information about subsequent fire is established [19, 21, 20]. Gelabert et al. [9] found dead fine fuel moisture and its anomalies to be the most influential predictor of human-caused ignition likelihood across Europe, testing generalisation by pooled fitting with per-site evaluation. What has not been tested is whether a classifier built on this class survives strict, label-free application to a region it never saw.

2.3. Spatial validation and transferability

Random cross-validation over spatially autocorrelated cells inflates skill estimates [33, 31], and spatially blocked designs are the standard remedy [39, 25]. The remedy is not uncontested: blocked schemes have been argued to introduce pessimistic bias of their own [43, 26, 6]. This paper uses spatial blocking throughout and reports every result at three block sizes, so that a reader can see which verdicts depend on the choice.

Blocking corrects the inflation that random folds produce within a region. It does not speak to performance on a fire the model has not seen, which Section 4.3 measures on identical cells. For that, the field’s instrument is the area of applicability and related predictor-space dissimilarity measures [23, 24, 18], which ask whether a target’s predictor values fall inside the training data’s envelope. Species distribution modelling has examined transferability far more systematically than fire has [46], and two studies there

report that geographic and environmental similarity do not reliably predict transfer success [41, 34], and the transferability of satellite-derived empirical
205 models is an active question in fire remote sensing itself [22].

2.4. *Dataset shift and label-free adaptation*

The canonical taxonomy separates covariate shift, where the predictor distribution moves but the predictor-response relationship holds, from concept shift, where the relationship itself changes [27]. Covariate shift is in
210 principle correctable without target labels, by per-region standardisation or by covariance alignment such as CORAL [37], and domain adaptation has an established remote-sensing literature [38, 30]. Concept shift is not: no realignment of input distributions can repair a reversal in the sign of an association, because detecting it requires the labels being withheld. Shift
215 decomposition in applied remote sensing is not itself new [12]. We are not aware of a prior application of covariance alignment to fire susceptibility, fire occurrence or burned-area prediction.

2.5. *Cross-region generalisation of fire models*

Few studies test fire-model transfer directly. Podschwit et al. [32] re-
220 port, in a Peruvian case study, that meteorologically derived danger indices do not port cleanly between fire environments. WildfireGenome [16] runs a leave-one-county-out matrix across seven US counties and reports strong within-county performance with highly variable off-diagonal transfer, similar pairs transferring well and dissimilar pairs collapsing; its label, however, is
225 a principal-component composite of hazard *indicators* rather than observed

burned area, and it applies no adaptation. Xu et al. [45] argue that wildfire transfer conclusions depend strongly on evaluation design, a caution we address by fixing the protocol in a project log before the diagnostics were computed and by reporting every sensitivity axis; that log is not a formal pre-
230 registration, and Section 3.11 states what was fixed and when. Kondylatos et al. [15] provide Mesogeos, a 1 km Mediterranean datacube for data-driven wildfire modelling.

Evaluation extent and AUC. Species distribution modelling settled long ago that the area a model is evaluated over is not a neutral choice. Lobo
235 et al. [17] make it the fifth and, by their own ranking, most important reason to distrust AUC as a comparative measure: the extent of the modelled area governs how many easy absences enter the calculation, and therefore the score. VanDerWal et al. [40] show the same lever on the calibration side, that how far background points are drawn from known presences changes
240 the fitted model and its apparent accuracy, and Barve et al. [3] give the argument its general form as the accessible area. This literature is qualitative about magnitude, because magnitude is problem-specific: it establishes that the effect exists, is directional, and can be large, without fixing a number for any given application. Two things follow for the present paper. First, our
245 Contribution 1 is a measurement inside that framework rather than a new phenomenon, and Section 1 states it that way. Second, the wildfire literature has largely not imported the lesson: region-wide validation figures are reported as though they described performance at the fire, and we know of no wildfire study that holds the model fixed and varies only the evaluation cells.
250 The number we obtain, 0.143 ROC-AUC, is larger than the predictor-block

increments this literature reports, which is why we treat it as a reporting problem in Section 5.8 rather than a caveat.

The nearest neighbour, and the contrast the present paper draws.

Dimarco et al. [8] is the closest Mediterranean analogue: 500 m predictors harmonised across four countries, tree ensembles under spatial cross-validation, transfer tested both leave-one-country-out and as a full 4 x 4 matrix, with every transfer exceeding AUC 0.80 and bioclimatically similar countries scoring higher. They apply no domain adaptation.¹

Much of the design is shared — Mediterranean regions, 500 m cells, MCD64A1-derived targets, tree ensembles, spatially aware validation, an explicit transfer matrix — and two things differ. **The predictor class:** their model rests on attributes of a place, all spatially stationary, ours on the state of a surface in one season. **The response variable:** theirs is human-driven ignition, ours is burned area, and those are different quantities with different dominant controls. Read together the two results suggest that the relationship between domain similarity and transfer success may be predictor-class dependent, with Vesik et al. [41] and Rousseau and Betts [34] aligning from species distribution modelling while Dimarco et al. and WildfireGenome run against it. We present that as a live disagreement, and Section 5.7 states why our own data do not settle it.

Methods, where no land surface temperature or TVDI variable appears and the only temperature predictor is a static ERA5-Land seasonal climatology [28]. We follow their Methods, which we take to be authoritative, and

¹Their Results text refers to "LST anomalies", which is inconsistent with their own

note the discrepancy so that the comparison is transparent. Every statement
275 about their work was checked against the full text, read 2026-08-08.

3. Data and methods

3.1. Study regions and temporal windows

Five Mediterranean wildfire regions are analysed (Fig. C.1): Manavgat
2021 and Muğla 2021 in Türkiye, Bejís 2022 in Spain, North Evia 2021 in
280 Greece and Montiferru 2021 in Sardinia. Each is a place-based rectangular
area of interest in EPSG:4326, defined from place coverage rather than from
a fire perimeter, deliberately not clipped to it so that unburned cells around
each fire form the negative class, and not tuned on burned prevalence, gate
outcome or any model metric. **One AOI choice was label-informed and**
285 **is stated as such:** the North Evia box was extended after the legacy box was
found to carry an atypically high burned prevalence, the extended geometry
then being defined from place anchors, with the legacy variant kept as a
sensitivity arm (Appendix A). Section 4.4 shows that this framing decision is
consequential and Section 5.9(x) treats it as the design lesson of the paper.
290 A sixth region, Kozan 2023, is carried as a negative control and excluded by
the gate of Section 3.3.

Each region has two non-overlapping windows. The **predictor window**
closes the day before the **label window** opens, so no predictor observation
can post-date the first labelled burning. Window lengths vary between re-
295 gions with the event: 57 to 61 days for predictors and 35 to 59 days for labels,
counted inclusively. A four-year baseline of window-symmetric composites
precedes each predictor window and supplies the climatological reference for

the anomaly channels. Region identifiers, bounding boxes, window dates and baseline years are registered in `core/regions.py` and are reproduced here.

300 Region identifiers, bounding boxes, window dates and baseline years are registered in `core/regions.py` and reproduced in Appendix C, Table C1.

3.2. *Burned-area label and the ~ 500 m analysis grid*

Labels come from MODIS MCD64A1 Collection 6.1 [10] retrieved through Google Earth Engine [11], whose omission and commission characteristics [4] bound every model here. The analysis grid is **reconstructed rather than native**: the pipeline’s 30 m EPSG:4326 reference grid is partitioned into 17 x 17 blocks, giving a nominal 510 m cell that approximates rather than reproduces the MODIS cell, and is square in degrees but not on the ground. A cell’s burn date is the mode of its positive sub-pixel day-of-year values, tested against the region’s label window; the label never affects a cell’s eligibility for modelling. Two safeguards are recorded rather than assumed: cells burning inside a region’s own predictor window are removed, which ran for three of five regions, and burning in earlier years is screened for none. Appendix C.1 gives the full specification, including what follows from the grid’s shape.

315 3.3. *Burned-landcover admissibility gate*

Before any modelling each region passes a gate asking what fraction of its burned cells is dominated by natural vegetation, using ESA WorldCover classes on the same cells. A region is admitted at 0.50 with at least 30 burned cells, and rejected as a cropland-dominated control at 0.50 cropland. 320 The purpose is to separate burned area produced by natural-fuel combustion

from post-harvest stubble burning, which MCD64A1 does not distinguish. Verdicts are in Section 4.1; the full rule is in Appendix C.1.

3.4. Predictor variables

Two feature sets are used. The **baseline** is terrain, fuel and green-
325 ness: elevation, slope, dominant land cover and a predictor-window median
vegetation-index composite, all static or near-static over the timescale at
which fire danger varies. The **thermal** set adds six pre-fire channels: current
land surface temperature, its anomaly against a four-year window-symmetric
baseline at the same cell, the Temperature-Vegetation Dryness Index and its
330 difference against that baseline, and two coordinate-informed products, a
downscaled and a fused surface temperature. The two internally differenced
channels are the ones constructed to isolate the dynamic anomaly. TVDI's
wet and dry edges are percentiles of the values a given area and window
happen to contain, so it is not portable as a physical quantity independently
335 of any concept shift; Section 5.9 carries the consequence. Provenance, com-
positing and the downscaler are in Appendix C.

3.5. Cell aggregation, validity and analysis populations

Cell-level values are means over the ~ 510 m cell, computed from valid
30 m pixels only, with the valid fraction recorded. A cell is **valid_for_**
340 **modeling** when it meets a 30 % valid-pixel floor and no thermal channel
is missing. Two populations are then defined. The **primary** population
is natural vegetation, cells whose combined tree, shrub and grass fraction
reaches 0.50, which excludes cropland from every burnable mask. The **sec-**
ondary population is all valid cells, reported throughout as a sensitivity.

345 Every headline result is given for both.

3.6. *Feature sets and classifier*

Two nested feature sets are compared. The **baseline** set is NDVI, elevation, slope and land cover. The **thermal** set is the baseline plus the six thermal channels. Land cover is one-hot encoded and numeric features
350 are standardised inside the fitting pipeline. The classifier is a random forest [5] with 300 trees, unlimited depth, `min_samples_leaf = 3`, balanced class weights and `random_state = 42`, identical for every region, population, feature set and transfer direction, so that no comparison in this paper is confounded by a model choice.

355 3.7. *Spatial-block cross-validation and bootstrap uncertainty*

Cross-validation is spatially blocked. Blocks are squares of the analysis grid formed as `row_500m // B` by `col_500m // B`, and folds are drawn over whole blocks, so no block is split between training and test. Five folds are used. The block size B is reported at 2, 10 and 20 cells, about 1, 5 and 10
360 km, for the within-region analysis.

Uncertainty is a spatial-block bootstrap: blocks are resampled with replacement, 1000 replicates, seed 42, with 2.5 and 97.5 percentiles as the interval. Because it is blocks that are resampled, what bounds an interval's reliability is the number of blocks carrying at least one burned cell, and those
365 counts are reported alongside the intervals. Where a verdict rests on too few such blocks it is stated as indicative rather than as an interval.

3.8. Cross-region transfer protocol

For each ordered pair of regions the model is fitted on the source population and applied to the target population with **no target label used**
370 **at any point**: no refitting, no threshold selection, no calibration. Both the baseline and the thermal feature set are transferred, so that the thermal block’s contribution is a paired difference within a direction rather than a comparison across directions. All twenty ordered directions among the five regions are computed.

375 3.9. Label-blind domain adaptation

Two label-free remedies are tested on every direction, in both feature sets.

Region-wise z-score. Each region’s numeric features are standardised using its own statistics, source statistics from source data and target statistics from target data, never pooled. The classifier is refitted on the z-scored
380 source and applied to the z-scored target. This removes first- and second-order marginal offsets.

CORAL after region-wise z-score. Starting from the z-scored features, the source covariance is aligned to the target covariance by the standard whitening-recolouring map [37], with regularisation $\lambda = 10^{-5}$ and eigen-
385 values floored at 10^{-12} . Critically **the transform is applied to the source only**; the target is left exactly as it is. The classifier is refitted on the aligned source. Neither variant sees a target label, and both are verified label-blind at run time.

λ sensitivity was assessed over nine values from 0 to 10^{-1} on four di-
390 rections in both feature families, moving transfer AUC by at most 0.014 in

any direction and 0.008 within the thermal family. The canonical $\lambda = 1$ lies outside the released sweep and is reported separately in Appendix A(b).

3.10. Transfer-gap decomposition and the concept-shift diagnostic

For each direction the gap between the target’s own within-region skill
395 and the raw transfer result is split in two. One part is what the best label-free adaptation recovers, and the other is what it does not. The recovered fraction is $(\text{adapted} - \text{raw}) / (\text{within} - \text{raw})$, signed and unclipped, with its interval from the same paired bootstrap; it bounds what covariate-level correction can achieve. Section 4.3 shows the remainder should not be read
400 as a conditional residual, because much of it is incurred inside a single region (Appendix A(j)).

The mechanism is diagnosed by **signed univariate association**. For each numeric predictor the raw ROC-AUC of that predictor against **burned** is computed in each region and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a
405 value below 0.5 is read as a direction rather than as weakness. A reversal is called bootstrap-supported only when the two regions’ point estimates fall on opposite sides of 0.5 **and each region’s own interval excludes 0.5**, under the same 10-cell spatial-block bootstrap. That is stricter than requiring the two regions’ intervals to be disjoint: a feature whose intervals are disjoint
410 but one of which straddles 0.5 has not been shown to point anywhere in that region, so it is recorded as a point reversal only. Appendix B states the rule again beside the counts, and `conditional_similarity_transfer.json` carries it as machine-readable metadata.

3.11. *Transferability diagnostics versus transfer*

415 Twenty candidate diagnostics from five families are computed for every region pair and rank-correlated (Spearman) against observed raw transfer AUC, under one bootstrap framework that resamples unordered region pairs with both of their ordered directions travelling together. The families are marginal predictor-space measures $P(x)$, including area-of-applicability dis-
420 similarity, climatic and geographic distance and a learned domain classifier; burned-niche overlap $P(x|y=1)$; burn-pattern regime distance $P(y)$; and conditional direction agreement $P(y|x)$. No family-wise error control is claimed, and the number of variants computed within each family is reported with the result. Appendix C.2 gives the full specification of each diagnostic and
425 of the bootstrap.

3.12. *Interventions*

Pooled multi-region training. Leave-one-region-out: the model is trained on the pooled primary populations of the other four regions and evaluated on the held-out region, with folds blocked as before. This asks
430 whether pooling recovers what single-source transfer loses.

Removal of direction-reversing features. The two predictors whose signed association reverses between regions with bootstrap support **on the frames as drawn** are **elevation_mean** (Manavgat against Bejís and against Muğla) and **lst_anomaly_mean** (Bejís against Evia). Section 4.4 with-
435 draws that support under an equalised frame, so this selection rule is frame-dependent and the arm below is reported as a measurement made under the original protocol rather than as a consequence of an established reversal. They are dropped and everything is refitted, within-region and across every

direction, and the two are also dropped singly so the cost can be attributed.

440 This measures what the removal costs locally and what it returns on transfer on the same footing. Note that the two features are selected by the same reversal analysis against which the result is then read, so both quantities are post-selection estimates and no correction for that selection is applied.

3.13. Controls on the transfer path

445 Four evaluations establish what the transfer arms are measuring. All use the transfer protocol of Section 3.8 unchanged — fit on the source cells, apply to the target cells, no refit, no recalibration, no threshold selection — and all were verified to reproduce the frozen cross-region AUCs before being used. They are a **within-region half-split** (the modelled cells cut in two at the median of a grid axis, both axes and both directions, a split discarded when
450 either half is single-class); **leave-one-scar-out** (each burned component of at least 50 cells, dilated by a 2 km buffer, withheld from training and used as the target); a **foreign-region evaluation** of the same held-out scar areas; and the **same blocked model restricted** to those areas, which isolates the
455 evaluation region from the training regime. Scars are dilated by buffers of 2, 5 and 10 km, with 2 km reported as primary; the minimum-component rule, and the per-split and per-scar positive counts, which are unequal and bear on the interpretation, are in Appendix A(i).

3.14. Leakage control

460 An explicit forbidden-column set is enforced at every model fit. Coordinates (`lon`, `lat`, `row`, `col` and their normalised forms), every burn-date and label-provenance column, and the agreement fraction are excluded from all

feature sets, and the enforcement runs as an assertion rather than a convention. The natural-vegetation mask is used only to define the population,
465 never as a predictor. Spatial blocking prevents a cell from sharing a fold with its own neighbours.

Reproducibility. All randomness uses seed 42 and the bootstrap uses 1000 replicates throughout. The transfer and adaptation analysis runs in an environment separate from the upstream pipeline’s. Every within-region
470 model was therefore refitted there and compared against the frozen upstream output, and the independently implemented adaptation was compared against the pipeline’s own. The within-region comparisons agree exactly and the twenty transfer directions to within 1.6×10^{-7} . All numbers here were produced under scikit-learn 1.9.0 or verified against it; the im-
475 plementation tolerance that applies if the version is not fixed is stated in Section 5.9(vi), and the companion paper reports the version sensitivity and the reproduction check in full.

Sensitivity analyses. Every headline result is repeated across two analysis populations, three spatial-block sizes, the CORAL sweep, both feature
480 sets and four classifier capacities; where a conclusion depends on one of those choices the dependence is reported rather than resolved by choosing the favourable setting (Appendix A).

3.15. Same-geography event-to-event comparison (Muğla 2021 versus 2022)

Muğla admits a comparison in which place is held fixed and the event
485 varies: a second fire burned inside the identical AOI, on the identical analysis grid, eleven months after the first. Signed univariate AUCs are computed for both arms under the same 10-cell spatial-block bootstrap used elsewhere.

Season, year and population all differ between the arms, since the 2022 arm is defined by removing the 2021 scar, and Section 4.10 reports what that costs. Appendix C.3 gives the window dates, the population construction and the asymmetry audit.

4. Results

4.1. Study regions and the admissibility gate

All five candidate regions pass the burned-landcover gate as wildfire candidates and the negative control fails it as intended. Kozan 2023 returns a natural-vegetation fraction of 0.017 against the 0.50 threshold and is excluded from all modelling. The separation is not marginal: the five admitted regions carry fractions of 0.723 to 0.991, so the threshold falls in an empty interval rather than between neighbouring cases. That is consistent with the gate separating burned area produced by natural-fuel combustion from post-harvest stubble burning, which MCD64A1 does not distinguish, though one negative control cannot establish it.

North Evia is analysed on an extended AOI. Relative to the legacy box, the extended box leaves the burned scar essentially unchanged while cutting overall prevalence from 0.361 to 0.122 and TSG prevalence from 0.676 to 0.287. The effect of that choice on transfer is reported in Appendix A(a).

4.2. Within-region: the thermal increment replicates in five regions

Adding the six thermal predictors to the baseline raises spatially blocked out-of-fold ROC-AUC in every region. The increment's bootstrap interval excludes zero in all five regions at 1 km and at 5 km blocking (Fig. C.3).

Those are the two scales this design supports as intervals. At 10 km the point estimates hold, from +0.048 to +0.154. They rest on 6 to 33 positive-carrying blocks, so they are reported as indicative rather than as intervals. The table note gives the reason.

515 *Table note (resampling units).* The bootstrap resamples spatial blocks, so what bounds an interval’s reliability is the number of blocks that carry at least one burned cell. Those counts fall quickly as blocks coarsen. At 2 cells they are 235 (Manavgat), 302 (Bejís), 843 (Muğla), 716 (Evia) and 192 (Montiferru), out of 5 439, 3 967, 11 316, 2 566 and 743 blocks. At 10
520 cells they are 28, 19, 70, 41 and 16, out of 237, 176, 576, 155 and 35. At 20 cells they are **12, 6, 33, 15 and 6**, out of 60, 48, 167, 50 and 12. No bootstrap replicate was invalid at any block size, with one exception. At 20 cells Bejís had 5 of 1000 replicates single-class. That is the symptom of the six positive-carrying blocks just reported, and a further reason to
525 read the row as indicative. An equal-tailed percentile interval built on six positive-carrying blocks has no meaningful coverage. Montiferru at 20 cells also feeds only 12 groups into a 5-fold grouped split, so its models train on about ten blocks each. **The 20-cell row of this table should be read as indicative rather than as an interval.** The 10-cell row is the coarsest
530 blocking this design supports properly, because every region there has 16 to 70 positive-carrying blocks. The increment holds at that scale in all five regions. Source: `paper/referee2_numbers.md`, block C, counted from the frozen per-cell prediction tables.

This within-region result is not itself novel. It is reported because the
535 transfer arms below are measured against it.

Table 1: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.7); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells $\approx$ 1/5/10 km.

Region	Block ($\approx$ scale)	Baseline AUC	Thermal AUC	Δ AUC	Δ AUC 95% CI
Manavgat 2021	2 ($\sim$ 1 km)	0.803	0.870	+0.067	[+0.055, +0.079]
	10 ($\sim$ 5 km)	0.748	0.797	+0.050	[+0.023, +0.077]
	20 ($\sim$ 10 km)	0.683	0.731	+0.048	[+0.014, +0.085]
Bejís 2022	2 ($\sim$ 1 km)	0.862	0.918	+0.056	[+0.048, +0.065]
	10 ($\sim$ 5 km)	0.779	0.825	+0.045	[+0.018, +0.069]
	20 ($\sim$ 10 km)	0.739	0.795	+0.057	[+0.031, +0.090]
Muğla 2021	2 ($\sim$ 1 km)	0.743	0.859	+0.116	[+0.106, +0.125]
	10 ($\sim$ 5 km)	0.698	0.777	+0.079	[+0.050, +0.105]
	20 ($\sim$ 10 km)	0.673	0.733	+0.061	[+0.030, +0.094]
North Evia 2021 (ext.)	2 ($\sim$ 1 km)	0.759	0.912	+0.153	[+0.142, +0.166]
	10 ($\sim$ 5 km)	0.716	0.864	+0.148	[+0.119, +0.183]
	20 ($\sim$ 10 km)	0.679	0.833	+0.154	[+0.124, +0.189]
Montiferu 2021	2 ($\sim$ 1 km)	0.781	0.883	+0.101	[+0.080, +0.125]
	10 ($\sim$ 5 km)	0.620	0.720	+0.100	[+0.017, +0.186]
	20 ($\sim$ 10 km)	0.555	0.681	+0.126	[+0.053, +0.228]

4.3. *What the evaluation frame is worth, within one region*

This section isolates the effect that Section 4.4 later applies between regions. It is reported first because it is measured on one model in one region with no transfer involved, so it cannot be explained by anything the transfer arms do.

Where the skill is lost, on a matched comparison. Four evaluations are reported. The last three are scored on **identical cells**, so they differ only in what the model was trained on, and the first is included to show why an unmatched comparison misleads. Each uses the transfer protocol unchanged: fit, then apply with no refit, no recalibration and no threshold selection.

The held-out unit is a burned connected component of at least 50 cells together with all cells within 2 km of it. What makes that a harder discrimination problem than a whole region is the composition of its negatives, not its burned fraction, and that can be shown rather than argued. Taking the same fitted model and the same out-of-fold predictions, and scoring them on a random sample of region cells drawn at the scar area’s own burned fraction, gives **0.782 against the region-wide 0.782**: matching the prevalence changes nothing, at $-0.000 [-0.003, +0.002]$ over the nine scars. Scoring the same predictions on the scar area itself gives 0.627, a fall of **+0.155 [+0.093, +0.217]**. The whole effect is therefore the negative pool. Every negative in a scar collar is fire-adjacent, sharing the terrain, land cover and synoptic conditions of the positives, whereas a region’s negatives include its easy far field. The burned fraction, 34 to 87 % against 3.8 to 28.7 % for a region, is a symptom of that construction, and ROC-AUC is in any case invariant to class balance at fixed class-conditional distributions.

Table 2: The four evaluations, scored on identical cells. Primary natural-vegetation population. Rows B, C and D are scored on the held-out scar area; row A is the whole region and is shown to make the mismatch visible. Means and Student t intervals are over the eight held-out scars.

Evaluation	Model trained on	Scored on	Mean AUC	95 % CI
A. Blocked cross-validation, 5 km	the region, scar included	the whole region	0.776	[0.738, 0.814]
B. Same blocked model, restricted	the region, scar included	the scar area	0.634	[0.552, 0.716]
C. Leave-one- scar-out	the region, scar withheld	the scar area	0.552	[0.501, 0.602]
D. Foreign region	another region, 306 to 2,802 km	the scar area	0.555	[0.495, 0.616]

All four rows are means over the **same eight scars**. A ninth burned component, Bejís, is excluded throughout: it is that region’s only component of any size, so holding it out leaves no usable source model and there is no row C for it. Rows A, B and D are reported here on the eight so that the
565 differences are paired, and the intervals are Student t over those eight, which is the resampling unit for this arm rather than the spatial-block bootstrap used elsewhere in the paper. Eight is a small number and the intervals are wide accordingly. Four of the eight scars are in Muğla and two in Montiferru, so they are not independent; row A in particular is a region-level quantity
570 repeated across the scars of a region, and its interval is pseudo-replicated and should not be read as coverage. The two differences are computed per scar and paired, which is what they are reported for.

The same model, scored two ways on the same region, differs by

0.143 AUC. Rows A and B are one model. The only change is which cells
 575 it is scored on: the whole region, or the burn scar and its 2 km collar. That
 change alone costs **0.143** [**+0.077**, **+0.208**] of the 0.225 [+0.157, +0.293]
 fall from A to C, about two thirds, and it is the size of the entire increment
 this literature usually reports. Both intervals are Student t over the eight
 scars, so the "two thirds" is a ratio of two estimates and is compatible with
 580 anything from roughly a third to nine tenths; the point of the comparison is
 the size of the numerator, not the precision of the fraction. A region-wide
 blocked figure is therefore an upper bound on what the same model achieves
 where the fire actually is, and the difference is not small enough to ignore.
 Any comparison of a scar-level result against a region-level reference inherits
 585 that, and the region-level reference is what a paper of this kind normally
 reports.

Neither withholding the fire nor moving 2,800 km has a measur-
able cost. B minus C, the fire-specific residual on identical cells, is **+0.082**
[−0.011, +0.175], which spans zero. C minus D, the effect of replacing
 590 a same-region model with one fitted between 306 and 2,802 km away, is
−0.003 [−0.075, +0.069], which also spans zero and is an order of mag-
 nitude smaller. The second is the cleaner result: on the same cells, a model
 fitted in another country does as well as one fitted in the same region with
 the scar withheld, and it survives restriction to Muğla, the only region where
 595 holding out one scar still leaves the source model properly trained, at 0.579
 against 0.597.

Row D is a mean over four foreign models, and the spread be-
hind it is large. Decomposing it gives **32** scar-by-source combinations

running from 0.374 to 0.724, with **ten below chance** and a mean spread of
600 0.171 across the four sources for a given scar (Appendix A, Table A3). The
pooled 0.555 is therefore not a statement that any foreign model does as well
as a same-region one, but that the *average* one does, and the variation it
averages over is exactly the pair-specific instability that the rest of Section 4
is about.

605 One confound must be stated. Holding out a region's only large scar
also removes most of its positives: the source model retains 11 positives in
Evia, 88 in Manavgat and 97 in Montiferru, against about 2,000 in Muğla,
which has four separate scars. Those four non-Muğla arms average 0.525
and the four Muğla arms 0.579. Three of the four are genuinely starved
610 — the fourth, Montiferru's second component, retains 472 source positives
— so row C mixes "the fire was withheld" with "almost all the positives
were withheld". Restricted to Muğla, C is 0.579 and D is 0.597, and the
C-to-D comparison still shows nothing. **With eight scars, three of them
starved, this design cannot establish a fire-specific residual, only
615 bound it at about 0.18.**

What this establishes is bounded rather than positive. Rows C and D
both sit close to chance, so the comparison between them has little dynamic
range, and neither the fire's identity nor the region boundary is shown to cost
anything here. What is measurable is the change of evaluation geometry
620 between rows A and B. Nor does it establish the fire event as the unit,
because the held-out patch is defined by the labels, so its identity cannot
be separated from its location. The within-region half-split, which does see
half of the target scar, returns 0.574 against leave-one-scar-out's 0.552, with

overlapping intervals.

625 **The result is robust to how the held-out patch is defined**, at 0.543 to 0.565 across minimum component sizes from 25 to 200 cells and under both 4- and 8-connectivity, and flat across buffers of 2, 5 and 10 km.

630 **The increment declines with the holdout, and is not established once the fire is withheld.** Contribution 1 is about the paired thermal-minus-baseline difference, so the same evaluations were run on it.

Evaluation	Thermal increment	Interval
Blocked cross-validation, per region	+0.056 to +0.153	every interval above zero
Within-region half-split	+0.027	positive in 13 of 18 splits
Leave-one-scar-out	+0.022	[−0.032, +0.077]
Cross-region, twenty directions	+0.004	[−0.028, +0.036]

635 The point estimate falls monotonically as the holdout hardens, and the last two intervals span zero. Under leave-one-scar-out the increment is positive in six of eight scars and averages +0.022, against a baseline of 0.529 and a thermal 0.552 that are both close to chance on that evaluation; restricted to Muğla it is +0.009 [−0.051, +0.070]. **The within-region increment of +0.056 to +0.153 is therefore substantially a property of interleaved holdout.** It is not shown to be zero on an unseen fire, and eight scars cannot show that; what this design establishes is that it is not established 640 there.

4.4. *The same effect between regions, applied to our own matrix*

Section 4.3's effect applies with equal force between regions, and this section reports that test before the transfer matrix rather than after it, because it changes what Sections 4.5 and 4.7 can claim. It withdraws nothing in Section 4.3, which is what the test is built from. Source: `aoi_frame_auc.csv`,
645 `aoi_frame_transfer.csv`, `collar_frame_bootstrap.csv`, `diagnostics_collar_frame.csv`, and the code deposited under `paper/code/`.

The five areas of interest are not comparable frames. Each region is a rectangle drawn around a fire, and the rectangles differ by an order of
650 magnitude in how much unburnt far field they enclose: the share of modelled cells lying beyond 10 km of any burned cell runs from **2.1 %** in Montiferru to **63.1 %** in Bejís, with median distances from 2.7 to 13.5 km (Appendix B, Table B6). The far field is not a neutral addition — in Manavgat the median elevation of modelled cells rises from 472 m within 5 km of the fire to 1,273
655 m at 20 to 50 km, against 512 m for the burned cells themselves.

Under an equalised frame the sign reversals of Section 4.7 do not survive. Restricting every region to cells within 10 km of any burned cell removes only far-field negatives; every burned cell is at distance zero and is retained at any radius, so the protection against choosing a flattering
660 radius is the sweep reported below, not the retention of positives.

On the equalised frame all five regions agree in sign on elevation, on LST and on TVDI, and both bootstrap-supported elevation reversals of Table B3 disappear. Applying this paper's own criterion from Section 3.10, which requires each region's own 10-cell block-bootstrap interval to exclude 0.5, **no**
665 **reversal remains bootstrap-supported in the collar frame** (`collar_`

frame_bootstrap.csv). Two features straddle 0.5 at the point estimate and neither is supported. `lst_anomaly_mean` runs 0.392 [0.324, 0.460] in Bejís against 0.584 [**0.497**, 0.669] in Evia, so the point estimates fall on opposite sides but Evia’s interval includes 0.5 by 0.003. `tvdi_difference_mean` straddles as well, at 0.509 [0.417, 0.597] in Muğla against 0.384 [0.288, 0.504] in Montiferru, and no region’s interval excludes 0.5. These are the two internally differenced channels, and an earlier draft of this section treated the first as uniquely informative because it carries no lapse-rate signal; that argument does not hold, because the second is equally decorrelated from elevation and behaves the same way. Both are point reversals under the strict criterion.

A weaker instrument does support the anomaly result. Table B3’s note commits this paper to a difference interval on the pair as the sharper test. Applied under the collar, four pairs have opposite-sided point estimates **and** a difference interval excluding zero, all on `lst_anomaly_mean` and three of the four involving Evia (Appendix A(1), Table A6). Three caveats keep it weak: nine of ninety feature-by-pair differences clear zero against about 4.5 expected under the null with no multiplicity correction, the nine features are effectively two to three dimensions, and Evia’s own interval-support status turns on 0.003. **The honest statement is that no reversal meets this paper’s strict criterion under the collar, and that the LST anomaly differs between regions on the weaker difference instrument.** Elevation under the collar is above 0.5 in all five regions but individually supported in only two.

The sign the five regions agree on is not the one the dryness

framing predicts. In every region a hotter pre-fire surface is associated with **less** burning, and the same holds for TVDI. It is neither a lapse-rate artefact nor greenness acting through fuel load: on mutual adjustment the surviving channel is LST, not NDVI. Two caveats bound it — in two re-
695 gions it is a residual spatial gradient that disappears when distance to the nearest burned cell is stratified inside the collar, and interval support is not uniform, so "all five agree" is a statement about point estimates. What we can defend is that the absolute thermal channels behave here as **static land-surface descriptors** rather than as a dryness index, and that the two
700 internally differenced channels carry no consistent cross-region direction at all. Compositing depth was not tested and remains an open alternative. Per-region values are in Appendix A(k); Section 5.2 states the consequence for the moisture-stress motivation of Section 1.2.

The same test determines what the diagnostics of Section 4.6
705 **can establish.** The only two candidates there with intervals excluding zero measure agreement in the sign of each predictor's association between source and target, counted over features interval-supported in both. Those are built out of exactly the signed AUCs this section has shown to be frame artefacts, and in Section 4.6 they are correlated against transfer measured on the same
710 unequal frames. Recomputing both sides under the 10 km collar:

The two fail differently and both fail. Once frames are equalised every region pair agrees in sign on every jointly supported feature, so the agreement fraction has no variance left. The supported cosine keeps a trace of variance — unanimous signs fix directions but not magnitudes — and simply stops
715 tracking transfer. This is not a marginal shift: features supported in both

Table 3: The diagnostics that ordered transfer, recomputed on an equalised frame. Spearman ρ against target ROC-AUC over the ordered directions in which each is defined. The first two rows are the two variants that cleared zero in Table B1; the third is the all-feature cosine, which did not and is shown for contrast. Both sides are recomputed here under one bootstrap setting, 1000 replicates, seed 42. Source `diagnostics_collar_frame.csv` and `collar_increment_and_cosine.csv`.

Diagnostic	Full frame	10 km collar
Sign-agreement fraction, supported features	$\rho = +0.86$ ($p = 0.0001$, $n = 14$)	1.0 in all 18 directions, variance exactly 0 — degenerate
Cosine, supported features	$\rho = +0.81$ ($p = 0.0005$, $n = 14$)	$\rho = -0.06$ ($p = 0.82$, $n = 18$), variance 0.00014
Cosine, all nine features	$\rho = +0.50$ ($p = 0.023$, $n = 20$)	$\rho = +0.12$ ($p = 0.61$, $n = 20$)

regions *rise* from 1.20 to 3.40 per direction, so the diagnostics are better determined and unanimous. The disagreements they were reading were the far fields.

Two scope statements belong with the table and are given in Appendix
720 A(p): the full-frame values here are not numerically the published ones, because the support test is itself bootstrap-dependent; and only these diagnostics were recomputed on the collar, so the other eighteen are unknown against the equalised transfer vector rather than shown to be null.

This settles what Contribution 3 can claim. It is not that conditional
725 similarity orders transfer where marginal similarity fails; it is that **no diagnostic tested here has been shown to order transfer**. The eighteen marginal, niche and regime candidates do not order it on the frames as drawn, and the two conditional variants that do stop doing so once the frames are comparable, because what they were reading is how the study rectangles
730 were drawn. The practitioner's position is therefore worse than the as-drawn analysis suggests, not better: there is no screen, and the apparent exception is an artefact. Section 4.6 reports its numbers as computed under the pre-registered protocol, which is what a reader following that protocol would obtain.

735 **The same-geography arm of Section 4.10 does not survive either, and it is the most extreme case in the cohort.** A fixed study area is not a fixed evaluation frame: its 2022 arm has 93.2 % of cells beyond 10 km of any burned cell against 55.3 % for 2021. Under the same collar the 2021 figure barely moves while the 2022 figure crosses to the same side of
740 0.5 with an interval covering chance, so **no bootstrap-supported sign**

reversal survives anywhere in this paper once evaluation frames are equalised, between regions or between two fires in one region. Values in Appendix A(m).

The within-region increment survives the same correction, remaining positive in all five regions on the equalised frame with a mean of +0.077 against +0.086 as drawn, and halving but staying positive in all five at a 5 km collar. The paper’s one surviving predictor-level positive claim therefore holds on the frame it argues is the correct one. Per-region values are in Appendix A(r).

The transfer matrix moves as well. Restricting source and target to the same collar:

The reference arm reproduces the frozen matrix, at 0.540 against Table B9’s 0.541 and 14 of 20 exactly, so this is measuring the same quantity. **The baseline control moves with it and must be restated on this frame:** the static baseline transfers at 0.593 against the thermal model’s 0.617, a paired difference of +0.023 rather than the +0.003 of the frame as drawn. The control still holds in kind — the static predictor class is not the portable one either — but the gap between them is eight times larger once frames are comparable, and Sections 1, 5 and 6 quote only the as-drawn pair. The above- and below-chance counts in this table are point counts. Under the same 10-cell block bootstrap used for Table B9, at 1000 replicates, the full frame gives nine directions above chance and four below with interval support, and the collar frame fifteen above and one below, so the headline movement is six to one at the point estimate and **four to one with interval support** (aoi_ frame_transfer.csv, which carries the per-direction bounds). The largest

Table 4: Cross-region transfer under equalised evaluation frames. Primary natural-vegetation population, thermal model, twenty ordered directions per row. Above/below chance are point counts; the supported counts use the same 10-cell (≈ 5 km) spatial-block bootstrap on the target as Table B9, 1000 replicates, seed 42. Per-direction bounds are in `aoi_frame_transfer.csv`.

Source frame	Target frame	Mean target AUC	Above chance	Below chance	Supported above / be- low	Ordered thermal delta
full	full (Table B9)	0.540	14 of 20	6	9 / 4	+0.003
full	10 km	0.575	17 of 20	3	11 / 1	+0.002
10 km	full	0.571	17 of 20	3	11 / 3	+0.014
10 km	10 km	0.617	19 of 20	1	15 / 1	+0.023
5 km	5 km	0.608	18 of 20	2	12 / 0	+0.014

movers are Bejís to Evia, 0.383 to 0.602, and Bejís to Manavgat, 0.440 to 0.601.

What this settles, and what it leaves standing. Five quantities reported in the sections that follow are properties of the frames rather than
770 of the predictor-burning relationship, and are identified as such where they appear: the count of six anti-predictive directions, which becomes one; the sign reversal of elevation, LST and TVDI as a mechanism; the sign-agreement diagnostic of Section 4.6; the same-geography arm of Section 4.10; and the paired thermal contribution, which is +0.004 as drawn and +0.023 equalised.
775 What the correction leaves standing is the central negative result, and its size must be stated on a matched comparison. Setting the equalised transfer mean of 0.617 against a within-region reference of about 0.87 would compare a 10 km-collar number with a full-rectangle one at 1 km blocking, which is the most generous reference in the paper and the very figure Section 4.3 argues
780 is an upper bound. Recomputing the within-region reference on the same frame and at the 5 km blocking this design defends (`matched_frame_gap.csv`, `paper/code/verify_matched_gap.py`):

Frame	Within-region (5 km blocking)	Mean transfer	Gap
full rectangle	0.798	0.540	0.258
10 km collar	0.772	0.617	0.155
5 km collar	0.737	0.608	0.129

Paired by target region, the collar shortfall is **+0.155** [**+0.094**, **+0.217**]
785 (Student t over the five target regions; per-region +0.086 Montiferru, +0.127 Manavgat, +0.161 Muğla, +0.196 Bejís, +0.206 Evia), which reproduces the point estimate above and excludes zero. The shortfall therefore survives

on every matched row, but it is **0.155 at the collar, not the 0.25 the unmatched comparison implies**, and it shrinks as the frame approaches
790 the fire — which is where a susceptibility surface is actually used.

This is Section 4.3’s effect acting between regions rather than within one, on frames whose fire-adjacent share ranges from 37 % to 98 %. Section 5.9(x) records the frame as a limitation of this cohort rather than of the method.

4.5. Cross-region transfer, and what label-free adaptation does to it

795 **Everything in this section is computed on the frames as drawn, and should be read against Section 4.4.** That section has already shown the frames are not comparable, that equalising them lifts mean transfer from 0.540 to 0.617 and leaves **one** direction below chance rather than six, and that the within-region reference used below is itself frame-dependent. The as-
800 drawn matrix is reported in full because it is what the pre-registered protocol yields and what a reader following that protocol would obtain; the corrections are not repeated at each number.

Raw transfer is heterogeneous and includes anti-predictive directions. Raw target AUC spans 0.326 to 0.686 (Fig. C.4). Twelve of 20
805 directions are above chance with CI support. Six are *below* chance with CI support: both directions of Manavgat to Bejís and of Manavgat to Muğla, plus Bejís to Evia and Evia to Bejís — of which only Manavgat to Bejís survives frame equalisation. Two intervals span 0.5. Even the best raw transfer, Evia to Manavgat at 0.686, stays far below that target’s own within-region
810 thermal performance of 0.870. Across all directions the raw deficit against the within-region reference is 0.184 to 0.592 AUC. That reference is a region-level blocked estimate, and Section 4.3 shows it is not matched to a transfer

evaluation.

Those counts belong to the 2-cell blocking of Table B9. At the more
815 conservative 10-cell blocking the same points give 9 above, 4 below and 7 un-
certain, and no direction changes side of the chance line (Section 4.9). Four
of the six below-chance directions keep their support there. Bejís to Man-
avgat and Manavgat to Muğla lose it and carry no verdict. The qualitative
statement is unchanged. The counts should not be read as exact.

820 **In precision terms the transfer is worse than the ROC fig-
ures suggest.** A susceptibility surface is used as a ranked area budget,
so precision-recall is the operational quantity. Across the twenty directions
the thermal model’s PR-AUC averages **0.156 against a no-skill baseline
of 0.136. Six of the twenty fall below their own no-skill baseline**
825 **at the point estimate, five of them with intervals entirely below it;**
the exception is Bejís to Manavgat, whose interval covers its baseline. Only
one direction exceeds twice its baseline. These are frame-as-drawn quantities
and the PR arm was not recomputed on the collar. Per-direction values are
in Appendix B, Table B4.

830 **The static baseline does not transfer either.** This is the control
for the reading the rest of the paper invites, and it constrains that reading
sharply. The same twenty directions were run with the terrain, fuel and
greenness baseline alone. The mean target AUC is **0.537**, against **0.541**
for the thermal model. Static attributes of a place are the class Dimarco
835 et al. transfer successfully, and are the class this paper’s framing treats as
portable. Here that class is itself barely above chance. The paired per-
direction contrast is given below and in `baseline_vs_thermal_transfer`.

csv. What matters here is that the transfer failure below is not specific to the dynamic block. A baseline that does not travel, plus pre-fire thermal
840 state, gives a model that does not travel.

The thermal block’s paired contribution to transfer, with its interval. Differencing the two matrices direction by direction gives a mean of **+0.004**, but the mean is not the informative statistic. The individual paired contributions span **−0.148 to +0.133**, twelve positive and eight negative,
845 so the spread is thirty times the mean and the sign is a property of the pair rather than of the block. A mean near zero here records cancellation, not consistent absence of effect. The directions are not independent, since each region appears in eight of the twenty, so the interval depends on the resampling unit; **all four units the design permits give the same an-**
850 **swer**, from $[-0.027, +0.034]$ treating directions as independent to $[-0.037, +0.046]$ jackknifing regions, and none propagates within-direction sampling variability (Appendix A(q)). The leave-one-region-out jackknife shows how little the mean is anchored: **dropping Evia alone reverses its sign.**

On the frames as drawn, six directions are below chance with
855 **interval support**, the sharpest at 0.326 $[0.305, 0.349]$. Section 4.4 has already shown that most of that count is a property of the frames: equalising them leaves **one** direction below chance, Manavgat to Bejís at 0.417 $[0.349, 0.488]$, supported at the 10 km collar though its interval covers chance at 5 km, at 0.459 $[0.372, 0.550]$. That one direction still needs a mechanism
860 acting on the direction of the relationship, because no account of merely lost skill produces a reliably reversed ranking, and Sections 4.6 to 4.10 pursue it. Everything reported below this point is computed on the frames as drawn,

so it should be read against Section 4.4 throughout. Per-split and per-scar detail is in Appendix A(i).

865 Per-direction values for all twenty directions, raw and under both adaptations, are in Appendix B, Table B9.

Label-blind adaptation compresses the matrix toward chance rather than repairing it. Under region-wise z-scoring the twenty directions span 0.431 to 0.630 and under CORAL 0.443 to 0.624, roughly half
870 the raw spread, with no adapted direction exceeding 0.631 against the unmatched within-region references of 0.859 to 0.918. Adaptation raises the failing directions and degrades most of those that already transferred. Taking the better of the two adaptations per direction, 14 of the 20 end closer to chance than they began and 6 end further from it; five of those six involve
875 Montiferru, the smallest and last-added region, and move upward, while the sixth is Manavgat→Muğla moving downward from 0.470 to 0.443. The 14 to 6 split should be read at the precision of limitation (ix) in Section 5.9, since Bejís→Manavgat is counted as compressed on a margin of 0.001.

The per-direction decomposition is in Appendix A(j), Table A5; two figures
880 ures from it matter here and are used below.

Against the right reference, adaptation is not failing (Fig. C.5). The three controls above give an achievable reference for a model applied to a fire it has not seen: 0.574 for the half-split, 0.552 for the leave-one-scar-out. CORAL, committed to in advance, averages **0.552** across the twenty direc-
885 tions against 0.541 raw, which is exactly that reference. Taking whichever of the two adaptations scores higher per direction gives 0.556, but that selection uses the target labels the protocol forbids, so it is an oracle upper

bound rather than an achievable result and is reported as one. What it does is regress the matrix towards that value: fourteen of twenty directions move
890 closer to chance, which harms the directions that already worked and helps the ones that did not. The verdict that follows is narrower than "alignment fails". Alignment cannot exceed what a model can achieve on an unseen fire, and it does not.

In the six directions where raw transfer was below chance, the best label-
895 free method recovers at most 34 % of the gap to the within-region reference, so the remaining unrecovered fraction is at least 0.66 everywhere. The half-split control above shows that the larger part of that remainder is already incurred inside the region, so it should not be read as a measure of concept shift. Seven directions show *negative* recovery, meaning adaptation moves
900 the score away from the reference; in six of those raw transfer was already above chance and adaptation destroyed that advantage. Label-free alignment therefore does not act as a repair mechanism.

4.6. Transferability diagnostics: what appears to order transfer, and why it does not

905 Twenty candidate diagnostics from five families were each rank-correlated with the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap.

Geographic separation does not order the matrix on either construction. Over all twenty directions the Spearman correlation between centroid separation and transfer AUC is -0.32 with an interval spanning zero; on the
910 twelve-direction common subset of Table 5 it is -0.24 , also spanning zero. The two nearest directions, Manavgat and Muğla at 306 km, are among the

Table 5: Transferability diagnostics versus raw thermal transfer, by family. Spearman ρ against raw transfer AUC with pair-based bootstrap 95 % CIs. Exp. is the sign expected if the diagnostic orders transfer. The member named is the one with the largest absolute correlation in its family, which is not always in the expected direction. All twenty individual diagnostics are in Appendix B, Table B1.

Family	Diagnostics	Exp.	Largest correlation in family	Sign as expected	Any CI excluding 0
P(y x) condi- tional	6	+	+0.84 [+0.58 , +0.88] agree- ment fraction, supported features	yes	yes, 2 of 6
P(x y=1) niche overlap	5	+	+0.24 [−0.45, +0.74] Schoener's D, 1-D mean	yes	no
P(x) marginal	6	−	−0.32 [−0.78, +0.33] domain- classifier AUC	yes	no
P(y) regime structure	2	−	+0.29 [−0.38, +0.74] ^{log} ₄₄ effective- N distance	no	no
geographic	1	−	−0.24 [−0.84,	yes	no

worst on the frames as drawn, while the 2,802 km pair returns 0.326 and 0.444, so the transfer mean of 0.541 is not the value at any one separation.

915 **Only two diagnostics have intervals excluding zero, and both are conditional.** The stronger is the sign-agreement fraction over interval-supported features, at $\rho = +0.84 [+0.58, +0.88]$. The cosine variant reaches +0.81. No marginal measure was shown to order the matrix. That includes area-of-applicability-style dissimilarity in predictor space, climatic distance
920 and geographic distance. The niche-overlap and regime families were not shown to order it either. The learned domain classifier is at ceiling, separating source from target at $\text{AUC} \geq 0.96$ for every pair. It always succeeds, which is why it carries no ordering information.

Four limits are stated with the result rather than after it. The first is
925 size. The index's tie structure caps the achievable Spearman at +0.861, so the observed +0.840 sits essentially on that ceiling. Its exact one-sided permutation p is 0.0060, the smallest this tie structure can produce, against a Bonferroni threshold of 0.0026 over the nineteen computed variants. No outcome of this diagnostic could have cleared family-wise correction on ten
930 effective pairs. The second limit is labels. Signed associations need burned labels in both regions, so the family that appears to work is not available before deployment, while the family that fails is.

The third limit is about selection, and it is the sharpest of the first three. The two rows that clear zero are the *supported-feature* variants. Their feature
935 subset is chosen by whether two regions' bootstrap intervals happen to be disjoint. That is a data-dependent selection, made on the same data, with no correction. The unselected counterparts over all nine features are $\rho =$

+0.50 $[-0.17, +0.83]$ for the cosine and $\rho = +0.18$ $[-0.40, +0.72]$ for the agreement count. Both span zero. The result lives in the selection step, and
940 is reported as such.

The families are compared on unequal samples, and equalising them does not change the ordering. The marginal, applicability, climatic and geographic rows sit on twelve directions, because those diagnostics exist only for the four-region subset. The supported-conditional rows
945 sit on sixteen and the rest on twenty. A reader may reasonably ask whether the marginal family’s failure is a statement about power rather than about diagnostics. Every row was therefore recomputed on the common twelve directions. The published values reproduce to 4.8×10^{-5} . The conditional rows still lead, at +0.87 $[+0.65, +0.88]$ and +0.85 $[+0.43, +0.88]$, and every
950 marginal row still spans zero. The ordering is not an artefact of unequal samples.

A fourth limit, established in Section 4.4, removes the result entirely. The sign-agreement index is built from the signed associations that Section 4.4 shows to be artefacts of the evaluation frames, and it was
955 correlated against transfer measured on those same frames. Recomputed on an equalised frame it is unanimous, taking the value 1.0 in every direction with no variance left to correlate, while the continuous cosine variant falls from $\rho = +0.50$ to +0.12. Everything in this section is therefore reported as what the original protocol yields, and the conclusion that survives is that **no**
960 **diagnostic tested here was shown to order transfer once the frames are comparable.**

4.7. The contrast pair: similarity is not sufficient

The clearest single view needs no ranking at all (Fig. C.8). Manavgat and Muğla lie in the same country and fire year, 306 km apart by centroid,
965 and their burned cells occupy the most similar environmental envelope of any pair in the matrix; Bejís and Montiferru occupy the least similar.

Per-quantity values for both pairs, on both frames, are in Appendix B, Table B10.

Two readings of this pair do not survive Section 4.4 and are not offered.
970 On the frames as drawn five of nine feature-response directions point opposite ways, elevation among them; under the collar that figure moves to 0.561 against 0.606, on the same side of 0.5. And transfer is below chance in both directions as drawn but above chance under the collar. **What survives is the ordinal contrast:** the most similar pair is among the weakest in
975 the matrix and the least similar among the stronger, while neither is the extreme — the weakest direction is Manavgat to Bejís at 0.417 and the strongest Muğla to Evia at 0.727. The claim this supports is that high envelope overlap does not buy transfer, not that it produces anti-prediction. The AoA shares are full-frame quantities.

980 Bejís and Montiferru sit at the opposite extreme. Their burned envelopes barely overlap, and they carry the most dissimilar values on every overlap measure. They transfer above chance in both directions. That half of the contrast rests on point estimates: neither direction carries a verdict at 5 km blocking, and the marginal applicability audit was never produced for
985 Montiferru. The claim is one of *sufficiency*. Similarity does not guarantee transfer, and dissimilarity does not preclude it. The claim rests on two

coexisting counterexamples, so it does not depend on the number of pairs available.

4.8. Interventions: pooling and feature removal

990 Two interventions were pre-registered. **Pooling** four regions into one training set does not recover what single-source transfer loses: every pooled target still falls short of its own within-region reference, and only Bejís is below chance with interval support. **Removing the two predictors whose signed association reverses** costs -0.081 of mean within-region AUC, 995 supported in every region and roughly three quarters of it attributable to elevation, a *baseline* terrain variable, and returns $+0.014$ $[-0.017, +0.045]$ on transfer, whose interval spans zero. A local cost is measured; no compensating transfer gain is, so no exchange between the two is demonstrated. Both predictors were chosen because they reverse, using the same data on 1000 which the two figures are then estimated, and no correction for that selection is applied; Section 4.4 additionally withdraws the support for the selection rule itself. Per-configuration values are in Appendix A(n).

4.9. Sensitivity analyses

Eight design choices were varied with everything else held fixed. They 1005 are the Evia AOI and its prevalence, the CORAL regularisation constant, the blocking scale, the closure date of the predictor window, the quality screening of the coarse thermal input, the contrast between the normalised and the absolute dryness channels, the removal of the coordinate-informed channels, and the capacity of the classifier. None changes a conclusion above.

1010 Two bound how the results should be read, so they are carried into the main text here.

Coarsening the blocks from 1 km to 5 km moves the **paired thermal-minus-baseline delta** verdicts from ten positive, seven negative and three uncertain to six, four and ten. The above-chance verdicts on the thermal
1015 arm itself, a different quantity, move from twelve, six and two to nine, four and seven. Support is removed from seven verdicts and added to none. The point estimates are unchanged, but that is an identity rather than a result. The blocking scale is the resampling unit, and it cannot move an estimate computed once over all target cells.

1020 Manavgat’s whole downstream chain was then rebuilt from a quality-screened MODIS input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed univariate association moves by more than +0.0003. This closes the one processing-artefact candidate for that region’s behaviour. Appendix A reports all eight arms, including one
1025 that tests a claim of Section 1.2 and does not uphold it.

4.10. The same geography, a second fire

Muğla burned twice, eleven months apart, on the same grid and through the same processing chain. The arm was designed to hold place fixed and vary only the fire, which would have separated regional concept shift from
1030 everything else that differs between study areas. On the frame as drawn it shows a bootstrap-supported elevation reversal between the two fires; Section 4.4 withdraws that, because the 2022 arm carries the largest far field in the cohort and under a 10 km collar the two arms fall on the same side of 0.5. The arm, its structural asymmetries and the direction of the bias they

1035 impose are in Appendix A(m).

4.11. Distance within a region, and why it does not reframe the result

One further arm measures how skill decays with distance inside a single region. A model is fitted on one half of a region and applied to the other, and target cells are binned by their distance from the training cells (`distance_`
1040 `curve.md`, `paper/code/distance_curve.py`). Means are unweighted over bins, whose positive counts range from 1 to 2,564, and the bins carry no intervals.

Separation from training cells	Bins	Mean target AUC
0 to 5 km	18	0.692
5 to 10 km	16	0.519
10 to 20 km	11	0.499
20 to 40 km	6	0.445
40 to 80 km	3	0.541
80 to 160 km	1	0.421
cross-region, 306 to 2,802 km	20	0.541

By 10 to 20 km inside a single region the model is already at
1045 **chance**. That is worth reporting on its own: it bounds how far a susceptibility surface of this kind can be carried from the cells it was fitted on, and it is consistent with Section 4.3, where withholding a scar and replacing the model with a foreign one cost nothing distinguishable.

This does not reframe the paper’s negative result, and a read-
1050 **ing that it does was considered and rejected.** That reading argued that because the twenty cross-region directions average 0.541, at or above the within-region plateau, they sit on the continuation of the curve and

the transfer failure needs no regional mechanism — a rule fixed in advance in `positive_control.md`. Two objections defeat it, and `scar_control.md` records the reframing as withdrawn in full. **The rule cannot fail.** Once the curve reaches the chance floor, any cross-region mean near 0.5 lies on its continuation by construction, so the comparison could not have come out otherwise and a test that cannot fail is not evidence. And **extrapolating an uninformative model does not produce reliably reversed ranking:** Manavgat to Bejís is below chance with interval support on the frame as drawn and at the 10 km collar, which is not what a model that has merely run out of skill returns.

Four further limits bound even the descriptive reading. The two distance ranges do not overlap — within-region separations span 2 to 86 km and cross-region separations start at 306 km — so any comparison across the gap is an extrapolation of the curve. The far bins are thin, six at 20 to 40 km and one beyond 80 km, so their means should not be read closely and the apparent rise at 40 to 80 km is not evidence of anything. The near bins are inflated by exactly the autocorrelation that blocked validation exists to remove, so **0.692 is an upper bound on near-field skill rather than an estimate of it.** And distance is not the only thing that changes with distance: this design separates distance from crossing a study-area boundary, but not from the land cover, terrain and fire history that covary with it.

What this arm therefore contributes is a length scale for the within-region decay, not an attribution. The unit that fails to transfer is not established by this design, and Section 4.3 says so directly.

4.12. What target labels cost: the recovery curve

Everything above measures a failure; this prices it. Label-free alignment does not close the residual gap, so the missing resource is information about the target that alignment cannot synthesise, and the direct way to supply it is target labels. A frozen labelled-budget diagnostic answers how many, for three regions across all six ordered directions, using one 10-cell (~ 5 km) spatial block as the unit of labelling effort and reading recovery against a matched target-only ceiling of 0.777 to 0.824.

Thirty-two labelled blocks recover **85 to 89 % of that ceiling in three of the six directions**, two of which started below chance; 51 to 57 % in two more, the directions where the conditional gap is widest; and 30 % in the sixth. That budget is 7 to 20 % of the target's natural-vegetation population, which at the 2,700 to 3,000 labelled cells those blocks carry (S1.3) and this grid's effective cell area of 0.199 to 0.208 km² (Appendix C) is roughly **540 to 620 km²**, so it is a real answer and not a cheap one. Two properties of the measurement bound it and are stated in the supplement rather than buried: at the top budget the selection pool is nearly exhausted, so the narrow upper-budget intervals reflect saturation rather than precision, and the labelled blocks are drawn from the event being predicted, which is not a resource available before that event burns. The protocol, the full curve and the limits are in Supplement S1.

Two limits belong with the number. At small budgets the same intervention damages the direction that already transfers best without any labels. And six directions in three regions cannot support a general label budget, so none is offered.

5. Discussion

5.1. Reading the three findings together

Section 1.4 states the three findings and Section 4 establishes them; this
1105 section argues from them rather than restating them. One relation between
them does need saying, because it is what makes the paper cohere: the first
finding is not a caveat attached to the other two, it is the instrument that sets
their size. Applied to our own matrix it withdrew three claims we had made,
and what it left standing is a shortfall in transferred skill, not a reversed
1110 relationship.

5.2. Why the thermal increment is real but local

The within-region increment and the transfer failure are measured at dif-
ferent separations, and Section 4.3 shows that most of the difference between
them is already present inside a single region. The increment should there-
1115 fore be read as local in a specific sense: it holds where held-out cells are
interleaved with training cells, and most of it is gone once they are not,
before the fire or the region changes. It is not an artefact to be explained
away — it replicates in five independent regions, survives coarsening of the
blocks to about 5 km with its interval intact, persists in the secondary pop-
1120 ulation, and survives a predictor window closed up to two weeks earlier —
but it is established under interleaved validation and not established beyond
it. The window-closure result is region-specific and should not be overstated:
the contribution strengthens in Bejís and Muğla, is flat in Montiferru, and
weakens monotonically in Evia, so what holds everywhere is survival,
1125 not improvement (Appendix A(d)).

The natural objection is that different Mediterranean regions are simply different systems, so that a predictor meaning one thing in one place and another elsewhere is a comparison of two systems rather than instability.

We designed the two-Muğla-events arm to answer that objection

1130 **and it does not answer it.** On the frames as drawn it looked decisive — region, grid and processing chain identical, static predictors identical cell by cell, and elevation’s association reversing between the two fires — but Sections 4.4 and 4.10 show it is instead the most extreme frame artefact in the cohort, and under the collar the two arms fall on the same side of 0.5.

1135 The objection therefore stands unanswered, and three further confounds were never resolved in any case. Season and year are confounded, and **that confound cannot be resolved in this study area**: the events sit 42 days apart in median burn day-of-year, neither year contains a second event at the other’s phase, and a calendar-matched arm would carry nine burned cells
1140 against a gate minimum of thirty. The population is not held fixed either, the 2022 arm being the 2021 arm with the 2021 scar removed. And the 2022 arm rests on eleven positive-carrying 5 km blocks against this design’s own floor of sixteen. With one fire per region everywhere else, and the one arm that held place fixed now withdrawn, **this cohort provides no evidence**
1145 **that the transfer shortfall is regional rather than event-specific**, and Section 5.9(v) is a substantive limit rather than a formality. Section 4.11 adds a length scale for the within-region decay — the model is already at chance by 10 to 20 km from its training cells — but explains why that cannot be turned into an attribution: once the curve reaches the chance floor, any cross-region
1150 mean near 0.5 lies on its continuation by construction.

One conclusion does survive from the other direction. On the frames as drawn the sharpest supported reversal belonged to elevation, a *baseline* terrain variable, and the static baseline transfers no better than the dynamic block. Whatever the shortfall is, it is not the thermal block’s peculiarity.

1155 *5.3. Why the similarity-based diagnostics fail*

A diagnostic built on distance in predictor space asks one question: do the target’s predictor values look like the training data’s? That is not the question that matters when the failure is conditional. A target region can sit well inside the training envelope while the relationship between those
1160 predictors and burning points the other way. Manavgat to Muğla is exactly that case, with 0.875 of target cells inside the weighted area of applicability and transfer among the weakest in the matrix. On the frames as drawn that pair is below chance in both directions; on the equalised frame it is 0.551 and 0.510, above chance but still among the weakest in the matrix, while the
1165 least similar pair reaches 0.669 and 0.624 (Section 4.7). The point does not need anti-prediction: high overlap does not buy transfer.

The same holds for the niche-overlap and regime families. At the point estimates, the pair with the highest burned-niche overlap in the matrix fails in both directions while the pair with the lowest transfers in both, so
1170 no monotone function of overlap can order the outcomes. The qualifier is not decorative: at the 5 km blocking this paper otherwise defends, neither Bejís↔Montiferru direction carries a verdict and Manavgat→Muğla loses its below-chance support, so this contrast is a statement about point estimates and is made as one. The domain classifier is at ceiling for every pair, which
1175 is informative in itself: the regions are trivially separable in predictor space

whether or not their transfer works, so separability carries no ordering information here.

5.4. *What the conditional diagnostic is, and what it is not*

The sign-agreement index was the paper's one diagnostic with an interval
1180 excluding zero, and Section 4.6 states three limits with the result: it sits
essentially on its own tie-structure ceiling, it could not have cleared family-
wise correction on ten effective pairs whatever it returned, and its feature
subset is chosen on the same data, its all-nine-feature counterparts spanning
zero. A fourth is that signed associations need burned labels in both regions,
1185 so it is a mechanism diagnosis rather than a pre-deployment screen.

A fifth limit removes it altogether. The index is built from signed asso-
ciations that Section 4.4 shows to be artefacts of the evaluation frames, and
recomputing it on an equalised frame leaves it unanimous and variance-free,
while the second variant that cleared zero loses its correlation entirely, from ρ
1190 $= +0.81$ to -0.06 . So the practical conclusion is not "use this index instead",
and it is not even "the index works but needs labels". It is that **none of
the twenty candidates was shown to order transfer**, and the two that
appeared to were reading how the study rectangles were drawn. That is a
worse position for a practitioner than Section 4.6 alone suggests, and it is
1195 the honest one. As always these are nulls on ten effective pairs: not shown
to order transfer, rather than shown incapable of it.

5.5. *What the two interventions do and do not show*

Both interventions show the same shape. Pooling four regions never beats
the best single-source transfer for any target and stays 0.28 to 0.50 below

1200 the within-region ceiling, so aggregation does not manufacture the missing
conditional information. Removing the reversing predictors costs -0.081 of
within-region skill with interval support in every region, and changes mean
transfer by $+0.014$ $[-0.017, +0.045]$, which is not distinguishable from zero.

That pair of numbers is easy to read as an exchange, and it is not one.
1205 The thermal block's own contribution to transfer is $+0.004$, with an interval
spanning zero. Removing the reversing predictors returns $+0.014$, with an
interval spanning zero. Both arms are null on the portability axis. What
the interventions measure is a local cost and no compensating transfer gain.
That is not a conservation law, and it is not a rate at which local skill can
1210 be sold for portability. No such rate is estimated here.

5.6. The regime hypothesis, reported as it happened

A regime-structure explanation was stated in advance and the data con-
firmed the null. The regime-distance correlation has the wrong sign at the
point estimate, the most regime-similar pair fails in both directions, and the
1215 most regime-different pair transfers above chance. One mundane explanation
can be set aside: cutting Muğla, by far the largest population, to Manav-
gat's cell count and separately to its positive count leaves Muğla's transfer
behaviour inside the subsampling range in both roles. The error was in the
hypothesised grouping, not in the data, and with ten pairs this cannot refute
1220 regime typology [2] in general.

5.7. The empirical contrast with Dimarco et al.

Dimarco et al. [8] transfer successfully across a comparable Mediterranean
design and we do not, and the two results are not in conflict. Their predictors

are attributes of a place, ours describe the state of a surface in one season,
1225 which suggests that the relationship between domain similarity and transfer
success is predictor-class dependent.

That reading is one of at least two, and we cannot separate them here.
Their response variable is human-driven **ignition**; ours is burned **area**. Ig-
nition likelihood is dominated by human access and activity, stable and
1230 near-universal drivers, whereas burned extent is dominated by spread. A
predictor-class explanation and a response-variable explanation are there-
fore confounded in this comparison. Our own data speak against a simple
predictor-class reading in any case: the static baseline transfers at a mean
of 0.537 here, so within this cohort the place-attribute class does not travel
1235 either.

5.8. Implications

For practice, one number should be stated before any other. In precision-
recall terms, which is how a susceptibility surface is actually used, transferred
models average a PR-AUC of 0.156 against a no-skill baseline of 0.136, and
1240 six of twenty directions fall below their own baseline. Whatever the ROC
figures suggest, a model moved to a region it was not fitted in does not
usefully rank burned cells there.

This paper also supports one concrete change in what is reported. Along-
side a spatially blocked within-region figure, report skill on a held-out burn
scar and its surroundings, or on a held-out fire event. On these five regions
1245 the two differ by about 0.14 ROC-AUC on the same model, which is the size
of the effect such papers usually claim, so a blocked figure alone should be
read as an upper bound. Transfer skill likewise has to be *measured* rather

than inferred from similarity, and the pre-deployment diagnostics currently
1250 used for that inference did not order it here. Where a model must be moved,
the resource that closes the gap is target labels: thirty-two labelled 5 km
blocks recover 85 to 89 % of the target’s matched ceiling in three of six direc-
tions, and 30 to 57 % in the rest. That is a real answer and not a cheap one,
being 7 to 20 % of the target’s natural-vegetation population, and at small
1255 budgets the same intervention damages the direction that already transfers
best.

For method development, the results bound what unsupervised alignment
can be asked to do. Two label-free methods applied carefully moved fourteen
of twenty directions towards chance, and five of the six they moved upward
1260 involve Montiferru, the smallest region; the sixth, Manavgat to Muğla, moves
downward. The committed-in-advance CORAL arm reaches a mean of 0.552,
and taking whichever of the two methods scores better per direction reaches
0.556 — but that selection uses the target labels the protocol forbids, so it is
an oracle upper bound rather than an achievable result (Section 4.5). Even
1265 the oracle only reaches the reference a model can reach on an unseen scar at
all (Section 4.3), so they are not failing far below an achievable target; they
are regressing the matrix onto it, which costs the directions that already
worked. What alignment cannot do is exceed that reference, and a sign
reversal is not a distribution mismatch that realigning inputs would repair.

1270 5.9. Limitations

The transfer failure is a finding, not a limitation. The limitations are the
boundaries on how far it generalises.

(i) **No meteorological covariates** enter the models, so we cannot say how local skill and portability would change with weather.

1275 (ii) **The same-geography comparison covers one region only**, and there year, seasonal phase and population are confounded; **that confound cannot be resolved in this study area** (Section 5.2).

(iii) **All labels derive from a single burned-area product**, MCD64A1, and no second product covers 2021 and 2022 at this resolution.

1280 (iv) **Evia remains the most imbalance-atypical population** even after the AOI extension.

(v) **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology and, since the one arm that held place fixed is withdrawn, this design cannot attribute the shortfall to region rather
1285 than to event.

(vi) **Cross-region point estimates carry an implementation tolerance** of roughly ± 0.02 to 0.03 if the library version is not pinned.

(vii) **The diagnostic correlations rest on an effective sample of ten region pairs**, and both the successes and the failures of Section 4.7
1290 should be read at that power.

(viii) **Manavgat’s atypical transfer behaviour remains unexplained**; three candidates were tested and none survives (Appendix C).

(ix) **The interval-support counts are less stable than the point estimates behind them**: the published split turns on a lower bound of
1295 -0.00045 , so the sign pattern is stable and the counts are not.

(x) **The five areas of interest are not comparable frames, and this cohort cannot fully repair it** (Section 4.4). The collar radius is itself

a choice and 5 km and 10 km do not agree exactly (0.608 against 0.617), and the frames were fixed upstream in **repo/**, so we can restrict them but
1300 not extend them. Any future cohort should fix the frame by an explicit accessible-area rule [3] before any predictor is computed; we treat that as the main design lesson of this paper.

(xi) **One classifier family.** The headline numbers use a random forest with unlimited depth, the configuration most able to encode local structure
1305 and least able to extrapolate. Appendix A(h) shows the transfer result is not an artefact of that choice: three further estimators, including a penalised linear one, all land between 0.510 and 0.556 and all place fourteen of twenty directions above chance. Those are point estimates without intervals, so the ordering among them is not claimed as a result. Other model families were
1310 not tried, and a different inductive bias might behave differently, but within this family the negative result is a property of the predictors rather than of an unregularised estimator.

6. Conclusions

Where a fire model is scored decides what it appears to know. Holding the
1315 model, its predictors and its fitting fixed and changing only which cells are scored, moving from a whole study region to the burn scar and its 2 km collar costs **0.143 ROC-AUC**. A control isolates the cause as the composition of the negative pool rather than class balance. The effect is the size of the predictor-block increments this literature publishes as findings, so a region-
1320 wide validation figure should be read as an upper bound on what a model achieves where fire actually occurs, and reported as one.

That correction is not only other people’s problem. Applied to our own five-region matrix it withdraws five of our claims. The study areas enclose very unequal far fields, and equalising them to a 10 km collar lifts mean transfer from 0.540 to 0.617, reduces the directions below chance, and leaves
1325 no sign reversal supported under our own criterion — including the one arm that held place fixed, two fires in the same study area eleven months apart, which we had wrongly exempted because its geography was constant while its evaluation frame was not. What had looked like a reversed predictor-burning
1330 relationship was mostly a statement about how five rectangles were drawn.

What survives is the central result. Pre-fire thermal dryness adds a real and repeatable within-region increment in all five regions, surviving spatial blocking at about 5 km and a predictor window closed up to two weeks before the first labelled burning. It does not travel: on matched frames and matched
1335 blocking, equalised transfer of 0.617 falls 0.155 short of the within-region reference, the paired cross-region contribution spans zero with a sign that varies by pair, and the static baseline transfers no better than the dynamic one on either frame, so this is not a peculiarity of thermal predictors. The shortfall also cannot be anticipated: none of twenty diagnostics from five
1340 families was shown to order the matrix, and the two that appeared to do so require target labels on both sides and stop ordering transfer once the frames are comparable. Label-free alignment compresses most directions towards chance rather than repairing them, and supervised recalibration works but is not cheap.

1345 Two limits should be read with all of this. The agreed thermal direction is the opposite of the one the dryness framing predicts — hotter pre-fire

surfaces burned less in every region, and holding greenness does not remove it while holding temperature reverses greenness in two regions — so on this cohort the absolute channels behave as static land-surface descriptors rather than as a dryness index. And with one fire per region, and the single arm that held place fixed now withdrawn, this design measures a transfer shortfall but cannot attribute it to region rather than to event.

Two practical consequences follow. Report local skill and portability separately, since a predictor block worth a large within-region increment may contribute nothing distinguishable from zero across regions. And fix the evaluation frame by an explicit accessible-area rule before any predictor is computed, because a frame chosen for convenience can manufacture both a transfer failure and a mechanism to explain it. Three extensions are left for future work: temporal transfer, meteorological covariates, and physically normalised dryness variables.

Appendix A. Appendix A. Sensitivity analyses

Each arm below varies one design choice and leaves everything else fixed. None changes a conclusion in the main text. They are reported so that a reader can see which choices were tested, and what each was worth. Arm (f) is the exception in one respect: it is not a robustness check but a test of a claim the introduction makes, and the claim is not upheld.

(a) Evia AOI and prevalence. The raw transfer arms were repeated with the legacy, high-prevalence Evia box. Every qualitative conclusion is unchanged. Thermal raw transfer AUCs move by up to 0.07, the largest being Evia to Bejís at 0.378 to 0.448, and no direction changes side of the

chance line.

(b) **CORAL regularisation, including the canonical $\lambda = 1$.** Over nine λ values from 0 to 10^{-1} on four directions, CORAL transfer AUC moves by at most 0.014 within any direction and 0.008 within the thermal family.

1375 The canonical $\lambda = 1$ of the cited method lies outside that sweep and was computed separately on the same four directions. It changes the thermal mean from 0.519 to 0.522, a shift of +0.003 with a per-direction range of -0.005 to $+0.016$, and none of those four directions crosses the chance line. The $\lambda = 1$ values are 0.510, 0.444, 0.559 and 0.575 against 0.508, 0.445, 0.564
1380 and 0.559 at $\lambda = 0.1$.

Those four directions are Bejís–Muğla and Manavgat–Muğla, and they exclude the pair that moves. An earlier version of this appendix generalised from them to the whole matrix, which was an error: the Manavgat–Bejís pair is not in the sweep and it is the pair whose CORAL
1385 result is λ -dependent. A separate frozen artefact, `experiments/cross_region/step10/coral_lambda_sensitivity.csv`, carries it. Bejís to Manavgat gives 0.5571 [0.5292, 0.5857] at $\lambda = 10^{-5}$, 0.5644 at $\lambda = 10^{-3}$ and 0.5473 at $\lambda = 0.1$, all with intervals entirely above 0.5, but **0.4850 [0.4496, 0.5197] at $\lambda = 1$** , which crosses the chance line and loses its interval support.

1390 Manavgat to Bejís falls likewise, from 0.5108 to 0.4776 [0.4501, 0.5013].

The correct statement is therefore narrower. CORAL-dependent conclusions are insensitive to λ over three orders of magnitude below 0.1; at the cited method’s canonical $\lambda = 1$ the one direction whose adapted interval sits entirely above chance loses that status. $\lambda = 10^{-5}$ is the minimally regularised
1395 choice and matches the reference implementation, but the dependence is real

and is reported rather than absorbed.

(c) **Blocking scale.** Recomputing the transfer quantities at 10-cell ($\approx$ 5 km) blocking from the frozen per-cell predictions widens the intervals and moves the verdict counts, from ten positive, seven negative and three uncertain at 1 km to six, four and ten at 5 km. Coarser blocking therefore removes
1400 support from seven verdicts and adds none. Across five seeds the counts run 5 to 6 positive, 3 to 4 negative and 10 to 11 uncertain, so they are not exact.

The point estimates are unchanged. That is an identity rather than a result, and it should not be offered as robustness. The blocking scale is
1405 the bootstrap *resampling unit*, and each point estimate is computed once over all target cells, so no choice of block size could have moved one. The comparison does establish two things. Every verdict that changes, changes towards "no verdict". And no direction crosses the chance line under the widened intervals. The counts are the fragile part of this paper. The sign
1410 pattern is not thereby shown to be robust. It is simply not tested by this variation.

(d) **Predictor-window closure.** The predictor window was closed 7 and 14 days earlier in all five regions. The thermal contribution stays positive and bootstrap-supported everywhere. The direction of the change is region-
1415 specific. It strengthens in Bejís, from 0.058 to 0.079 at 14 days, and in Muğla, from 0.115 to 0.128. It is flat in Montiferru. It weakens monotonically in Evia, from 0.156 to 0.149 to 0.135. What holds in every region is survival, not improvement. That is the claim made in Section 5.2.

(e) **Quality screening of the coarse thermal input.** Two of the
1420 five regions' MODIS inputs are quality-screened and three are not. The

split follows export date rather than design, and it induces an elevation-correlated change at the input, at $r = +0.615$ in Manavgat. Manavgat's entire downstream chain was rebuilt from a quality-screened input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C.

1425 No signed univariate association moves by more than +0.0003. Elevation is identical in both arms, at 0.374, the population is unchanged, and the within-region increment moves from $[+0.055, +0.079]$ to $[+0.054, +0.077]$. The reason is structural. Elevation is a DEM variable the screening cannot touch, and fusion falls back on the MODIS-derived surface across only 2.14
1430 percentage points of coverage. Details are in `paper/modis_qc_downstream_propagation.md`.

(f) **Normalised against absolute dryness channels.** Section 1.2 argues that an internally normalised index should be less exposed to absolute-temperature offsets between regions than raw land surface temperature. The
1435 thermal block contains both kinds, so the argument can be tested directly as a feature-set contrast.

One thing should be said before the result, because it makes the outcome less surprising than it might otherwise appear. The normalised channels are the two that hold their direction in the same-geography two-event
1440 comparison of Section 4.10, where the absolute channels move. That is a statement about one region across two fires. **Across regions it does not hold:** `1st_anomaly_mean` is itself one of the two predictors whose reversal is bootstrap-supported on the frames as drawn, reversing between Bejís and Evia (Section 4.4 withdraws that support under an equalised frame), which
1445 is why Section 4.8b drops it alongside elevation. Being internally normalised

protects a channel against the offset between two seasons in one place. It does not, on this evidence, protect it against a change of place.

Three feature sets were run over all twenty directions and all five within-region folds, with the classifier, population, folds and bootstrap held fixed.

1450 The harness aborts unless its reference configuration lands on the frozen exports. It did exactly: the maximum absolute difference between the reference arm and the frozen transfer AUCs is 0.000000 across all twenty directions.

Feature set	Mean transfer AUC	Directions > 0.5	Mean within-region AUC
baseline only	0.5371	16	0.7896
baseline + normalised anomalies	0.5442	15	0.8551
baseline + absolute surface state	0.5479	13	0.8564
all ten features (reference)	0.5414	14	0.8883

1455 **The advantage is not found.** Per direction, the normalised set minus the absolute set has a mean of -0.0037 . It is positive in 12 of 20 directions and runs from -0.099 to $+0.094$, so the spread is an order of magnitude larger than the difference. The normalised channels transfer very slightly worse at the mean, and nothing here supports a design rule favouring anomaly-referenced dryness for portability.

1460 Within region the two sub-blocks are indistinguishable, at 0.8551 and
0.8564, and each recovers most of the gap between the baseline’s 0.7896 and
the full model’s 0.8883. What separates them in sign stability does not
become transferable skill.

This arm strengthens the negative finding rather than softening it. Four
1465 differently constituted feature sets were tried. None clears a mean of 0.548
across regions, and all differ sharply within them. The baseline arm also
gives an independent confirmation of the control reported in Section 4.5: its
mean transfer of 0.5371 was recomputed here from the modelling datasets
and matches the 0.537 read from the frozen per-direction export. Details are
1470 in `paper/anomaly_only_transfer.md`.

The comparison is between two sub-blocks of one thermal set on one
cohort. It does not test normalised dryness indices in general, and it does
not test a normalisation fitted against a pooled multi-region reference rather
than each region’s own baseline years.

1475 **(h) Model capacity.** Every number in this paper comes from one ran-
dom forest with unlimited depth and `min_samples_leaf = 3`. That is the
configuration most able to encode local structure and least able to extrapo-
late, so a reader may reasonably ask whether the transfer failure is a property
of the predictors or of the estimator. Holding the model fixed is right for the
1480 internal comparisons and does not license the external claim, so three fur-
ther estimators were run over the same twenty directions and the same five
within-region folds.

Estimator	Within-region AUC	Within increment	Transfer AUC	Transfer increment	Above chance
Random for- est, depth un- lim- ited, leaf 3	0.888	+0.099	0.541	+0.004	14 of 20
Random for- est, depth 6, leaf 50	0.829	+0.055	0.556	−0.007	14 of 20
Random for- est, leaf 200	0.805	+0.047	0.550	−0.021	14 of 20
Penalised lo- gis- tic re- gres- sion	0.741	+0.045	0.510	−0.024	14 of 20

Regularisation does not rescue transfer. All four estimators land

1485 between 0.510 and 0.556, all four put exactly fourteen of twenty directions
above chance, and the linear model, the one built to extrapolate, transfers
worst. Within-region skill falls as capacity is reduced, from 0.888 to 0.741,
which is what regularisation is expected to cost; transfer does not rise to
meet it. The thermal block’s contribution to transfer is +0.004 under the
1490 canonical forest and **negative** under all three regularised alternatives, so the
block does not become portable when the model is made simpler.

The transfer failure is therefore a property of the predictors on this cohort,
not of an unregularised forest. Detail in `paper/model_capacity.json`.

(g) **The coordinate-informed channels.** `downscaled_1st_mean` and
1495 `fused_1st_mean` come from a per-region downscaling model whose own in-
puts include coordinates, which Section 3.14 names as the one route by which
a coordinate-derived surface re-enters a feature set that excludes coordinates.
A coordinate-smoothed surface is by construction locally informative and
non-portable, so if the within-region increment depended on it, that incre-
1500 ment would be this paper’s own result in miniature rather than a finding
about thermal dryness. The arm was therefore run here rather than de-
ferred.

Both channels were dropped and everything refitted, over all twenty direc-
tions and all five within-region folds. The reference configuration reproduces
1505 the frozen exports exactly, at 0.000000 against both the transfer and the
within-region references.

Region	Increment, full set	Increment without the two channels	Retained
Manavgat 2021	+0.067	+0.063	94 %
Bejís 2022	+0.056	+0.046	82 %
Muğla 2021	+0.116	+0.097	84 %
North Evia 2021	+0.153	+0.145	94 %
Montiferru 2021	+0.101	+0.105	103 %
Mean	+0.099	+0.091	92 %

The increment does not depend on them. It stays positive in every region and retains 82 % to 103 % of its size, rising slightly in one region. Mean transfer is likewise unchanged, at 0.545 without the two channels against 0.541 with them and 0.537 for the baseline alone. The local skill this paper reports is therefore not an artefact of a coordinate-smoothed surface.

This arm was computed independently here and reproduces the companion paper’s figure for the same quantity, 82 % to 103 %, from a separately written harness.

Appendix A.1. A(i). The four evaluations of Section 4.3, in full

Section 4.3 reports four evaluations of the same models as a ladder. The per-split and per-scar detail is here, so that the ladder can be checked without leaving the manuscript.

Table A1. Within-region half-split, every split. Source and target positive counts are given because they are unequal, which is the principal limit on this arm: a straight cut does not produce two exchangeable halves. Two splits are unusable because one half of Manavgat contains no burned cells.

Region	Axis	Direction	Source positives	Target positives	Thermal AUC	Baseline AUC
manavgat 2021	east-west	low to high	700	84	0.695	0.661
manavgat 2021	east-west	high to low	84	700	0.557	0.580
manavgat 2021	north-south	low to high	0	784	—	—
manavgat 2021	north-south	high to low	784	0	—	—
bejis 2022	east-west	low to high	640	460	0.750	0.784
bejis 2022	east-west	high to low	460	640	0.578	0.560
bejis 2022	north-south	low to high	357	743	0.713	0.662
bejis 2022	north-south	high to low	743	357	0.603	0.613
muğla 2021	east-west	low to high	1,449	1,462	0.448	0.378
muğla 2021	east-west	high to low	1,462	1,449	0.636	0.628
muğla 2021	north-south	low to high	774	2,137	0.533	0.503
muğla 2021	north-south	high to low	2,137	774	0.294	0.549
evia 2021 extended	east-west	low to high	2,052	612	0.589	0.506
evia 2021 extended	east-west	high to low	612	2,052	0.645	0.482
evia 2021 extended	north-south	low to high	2,564	100	0.649	0.513
evia 2021 extended	north-south	high to low	100	2,564	0.533	0.495
montiferru 2021	east-west	low to high	410	129	0.475	0.461
montiferru 2021	east-west	high to low	129	410	0.507	0.513
montiferru 2021	north-south	low to high	438	101	0.583	0.511
montiferru 2021	north-south	high to low	101	438	0.536	0.447

Table A2. Leave-one-scar-out at a 2 km buffer, every scar. Muğla and Montiferru are the only regions containing more than one burned component of at least 50 cells. Muğla is the only region where holding one out still leaves the source model properly trained across every arm; Montiferru’s two components are very unequal, so one of its arms retains 472 source positives and the other 97. Muğla’s four arms mean 0.579; the four arms in the other regions mean 0.525, and the pooled figure of 0.552 averages the two.

Region	Component	Source	Target positives	Target cells	AUC
		positives			
		left			
evia 2021 extended	1	11	2,653	3,059	0.465
manavgat 2021	1	88	696	1,135	0.592
montiferru 2021	1	97	442	758	0.584
montiferru 2021	5	472	67	195	0.458
mugla 2021	6	1,997	914	1,266	0.595
mugla 2021	1	2,173	738	1,244	0.561
mugla 2021	10	2,272	639	940	0.617
mugla 2021	8	2,363	548	954	0.542

1535 The evaluation populations of the two arms are not comparable with
each other or with the transfer targets. A held-out scar with its 2 km collar
contains only fire-adjacent negatives, against a whole target region’s inclusion
of its easy far field; the burned fractions, 34 to 87 % against 3.8 to 28.7 %,
are a symptom of that rather than the cause, since ROC-AUC is invariant to
1540 class balance at fixed class-conditional distributions. The scar arm therefore
asks for discrimination against the nearest and most similar negatives only,
while a transfer arm includes the whole easy far field. Section 4.3 states the
consequence: the last three rows of the ladder are not distinguishable by this
design.

1545 **Table A3. The foreign-region arm, decomposed by source.** Each
held-out scar area is scored with a model fitted on each of the other four
regions. Table 2’s row D is the mean over the **eight** scars that carry a row
C, that is 32 of the 36 combinations below; the nine-scar mean quoted in this
appendix is 0.559 against row D’s 0.555.

Target region	Scar	Mean over sources	Min	Max	Spread
Bejís 2022	1	0.589	0.521	0.631	0.110
North Evia 2021	1	0.590	0.374	0.704	0.329
Manavgat 2021	1	0.468	0.407	0.563	0.156
Montiferru 2021	1	0.504	0.457	0.546	0.089
Montiferru 2021	5	0.490	0.399	0.656	0.257
Muğla 2021	1	0.518	0.425	0.608	0.183
Muğla 2021	6	0.568	0.533	0.597	0.064
Muğla 2021	8	0.632	0.570	0.724	0.155
Muğla 2021	10	0.671	0.579	0.716	0.137

Over all 36 combinations here the mean is 0.559, the range 0.374 to 0.724, and ten fall below chance. Restricted to the 32 combinations behind row D the mean is 0.555 and the same ten fall below chance. The mean spread across the four sources for a single scar is 0.164 over the nine scars and 0.171 over the eight. Averaging over sources is what makes row D comparable with row C, which is fitted on one region; it is not a claim that the choice of foreign source is immaterial, and Section 4.3 states the distinction.

Table A4. Prevalence is not the cause of the evaluation-area effect. The same fitted model and the same out-of-fold predictions are scored three ways: on the whole region, on a random sample of region cells drawn at the scar area’s own burned fraction, and on the scar area. Twenty draws per scar, seed 42. Nine scars, since this control needs no leave-one-scar-out arm and Bejís therefore qualifies.

Target region	Scar	Burned fraction	A whole region	A'	B scar area
				prevalence- matched	
Manavgat 2021	1	0.61	0.797	0.796	0.589
Bejís 2022	1	0.66	0.824	0.825	0.575
Muğla 2021	6	0.72	0.777	0.772	0.564
Muğla 2021	1	0.59	0.777	0.779	0.654
Muğla 2021	10	0.68	0.777	0.779	0.673
Muğla 2021	8	0.57	0.777	0.782	0.761
North Evia 2021	1	0.87	0.864	0.866	0.745
Montiferru 2021	1	0.58	0.720	0.717	0.622
Montiferru 2021	5	0.34	0.720	0.723	0.461
Mean			0.782	0.782	0.627

1565 A minus A', the effect of prevalence alone, is **−0.000** [**−0.003**, **+0.002**].
A' minus B, the effect of replacing the region's negatives with fire-adjacent
ones, is **+0.155** [**+0.093**, **+0.217**]. The evaluation-area effect is entirely
the negative pool.

Appendix A.2. A(j). The transfer-gap decomposition, in full

1570 Section 4.5 states the two figures this table carries — at most 34 % of the
gap recovered in the directions that started below chance, and seven direc-
tions with negative recovery. The table is kept here rather than in the body
because Section 4.3 shows the within-region reference in its denominator is
not matched to a transfer evaluation, and Section 4.4 shows its raw column
1575 is frame-dependent, so the fractions should be read as within-protocol quan-
tities.

**Table A5. Transfer-gap decomposition (four-AOI set, 12 direc-
tions).** Within = target's within-region thermal AUC; best adapted = the

better of z-score/CORAL; recovered fraction = (adapted - raw)/(within
 1580 - raw), signed and unclipped, with paired bootstrap CI (1000 replicates).
 Montiferru directions are not part of this decomposition (per-pair absolute
 decompositions exist without fraction CIs). The status column asks whether
 the *adapted* value clears chance and uses the 2-cell adapted intervals of Ta-
 ble B9. The adapted arms were not recomputed at the coarser blocking of
 1585 Appendix A(c), which covers the raw arm and the paired delta only.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Manavgat→Bejís	0.918	0.326	0.511 (CORAL)	+0.31 [+0.28, +0.34]	recovery, chance not excluded
Bejís→Manavgat	0.870	0.444	0.555 (CORAL)	+0.26 [+0.18, +0.34]	recovery above chance
Muğla→Manavgat	0.870	0.401	0.560 (CORAL)	+0.34 [+0.28, +0.40]	recovery above chance
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	recovery above chance
Evia→Bejís	0.918	0.448	0.549 (z-score)	+0.22 [+0.16, +0.27]	recovery above chance
Manavgat→Muğla	0.859	0.470	0.443 (CORAL)	−0.07 [−0.12, −0.03]	negative recovery
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	−0.07 [−0.16, +0.01]	negative recovery
Manavgat→Evia	0.912	0.613	0.542 (z-score)	−0.23 [−0.32, −0.14]	negative recovery
Evia→Manavgat	0.870	0.686	0.527 (CORAL)	−0.86 [−1.18, −0.60]	negative recovery
Evia→Muğla	0.859	0.577	0.530 (CORAL)	−0.17 [−0.22, −0.11]	negative recovery
Muğla→Evia	0.912	0.653	0.563 (CORAL)	−0.35 [−0.43, −0.27]	negative recovery
Bejís→Muğla	0.859	0.618	0.518 (z-score)	−0.42 [−0.51, −0.34]	negative recovery

Appendix A.3. $A(k)$. The thermal sign, stratified

Section 4.4 reports that all five regions agree on a negative association between pre-fire surface temperature and burning, and that neither of the

1590 two obvious confounders explains it. The per-region values are here. Signed AUC against **burned** within the 10 km collar; stratified columns pool within-stratum concordance over deciles of the named variable, weighting each stratum by its positive-negative pair count. Source `matched_frame_gap.csv`, recomputable by `paper/code/verify_matched_gap.py`.

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Region	LST raw	LST within elevation	LST within NDVI	LST within distance	NDVI raw	NDVI within LST
Manavgat	0.386	0.402	0.440	0.505	0.621	0.542
Bejís	0.405	0.509	0.486	0.350	0.618	0.574
Muğla	0.332	0.363	0.404	0.367	0.652	0.547
Evia	0.286	0.327	0.279	0.319	0.663	0.380
Montiferru	0.376	0.392	0.368	0.485	0.582	0.405

Three readings follow. The LST sign survives stratification within elevation in four of five regions and within greenness in all five, despite $r(\text{LST}, \text{NDVI})$ reaching -0.92 , so it is neither a lapse-rate proxy nor an inverse-greenness proxy. It does *not* survive stratification within distance to the nearest burned cell in Manavgat or Montiferru, which are the two regions where the collar leaves the least residual gradient, so in those two the agreed sign is a weaker version of the same spatial effect the collar was introduced to remove. And the reciprocal adjustment runs one way only: NDVI reverses in two regions once temperature is held, while LST reverses in none once greenness is held.

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Appendix A.4. A(l). The LST anomaly under the difference instrument

Section 4.4 reports that no reversal meets this paper’s strict criterion once evaluation frames are equalised, and that the LST anomaly nonetheless differs between regions on the weaker instrument Table B3’s note commits

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the paper to. These are the four pairs that have opposite-sided point estimates and a difference interval excluding zero, all on `lst_anomaly_mean`. Signed AUC within the 10 km collar; the two regions are bootstrapped independently under the 10-cell spatial-block scheme and differenced, 1000 replicates, seed 42. Source `matched_frame_gap.csv`, recomputable by `paper/code/verify_matched_gap.py`.

Table A6. Between-region differences in the signed LST-anomaly association, 10 km collar.

Pair	AUC A	AUC B	Difference	95 % CI
Bejís vs Evia	0.392	0.584	−0.191	[−0.295, −0.083]
Evia vs Montiferru	0.584	0.400	+0.184	[+0.012, +0.321]
Manavgat vs Evia	0.462	0.584	−0.122	[−0.225, −0.020]
Bejís vs Muğla	0.392	0.507	−0.115	[−0.219, −0.002]

Nine of the ninety feature-by-pair differences clear zero, against about 4.5 expected at nominal 5 % under the null, and no multiplicity correction is applied; four of the nine are this feature and three of those involve Evia. The nine features are effectively two to three dimensions (Section 4.4), so the ninety comparisons are not independent either. This is reported as a weaker result than a reversal, not as a restored one.

Appendix A.5. $A(m)$. The same-geography event pair

Section 4.10 states the result and its withdrawal; the design, the two structural asymmetries that have no analogue in the twenty-direction matrix, and the direction of the bias they impose are here.

This arm was designed to hold place fixed and vary only the fire, which would have separated regional concept shift from everything that differs be-

tween study areas. Muğla burned twice, in 2021 and again eleven months later, on the same grid and through the same processing chain, and the 2022 arm is the 2021 population with the 2021 scar removed: 41,730 rows /
1635 2,911 burned for 2021 against 38,790 rows / 331 burned for 2022. **Positive-carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm,** and eleven is below the sixteen this design sets as its own floor, so the 2022 intervals are read as indicative exactly as the 20-cell row of Table 1 is. Full per-feature values are in Appendix B, Table B5.

1640 **On the frame as drawn, elevation reverses with bootstrap support.** In 2021 higher ground burned preferentially, at 0.611 [0.532, 0.690]; in 2022 lower ground did, at 0.296 [0.230, 0.355]. The intervals are disjoint and the difference is -0.317 [-0.414 , -0.220]: the same predictor, region and grid, and an association pointing the opposite way in two fires eleven
1645 months apart. The four absolute thermal channels change side as well.

Section 4.4 has already withdrawn it. The 2022 arm is one compact scar inside the whole Muğla box, so 93.2 % of its cells lie beyond 10 km of any burned cell against 55.3 % for the 2021 arm — the most extreme far field in the cohort. Under the same 10 km collar the 2021 figure barely moves, 0.611
1650 to 0.606 and still supported, while the 2022 figure moves from 0.297 to 0.565, onto the same side of 0.5 as 2021 with an interval covering chance. The arm shows the same artefact as the cross-region reversals rather than confirming them. We keep the section because the structural properties below have no analogue elsewhere in the paper, and because this arm is what motivated the
1655 frame test.

The two arms are not disjoint samples: they share 38,789 of the 2022

arm’s 38,790 cells, and elevation, slope and land cover are identical to the digit across all 73,098 grid cells, so only NDVI and the six thermal channels carry new information between them. And the removal is the target’s own
1660 positive class, so in the 2022-to-2021 direction not one target positive is in the source training population while 38,789 of 38,819 target negatives are, and membership of the source training set alone separates the 2021 target’s classes at ROC-AUC 0.9996. **We therefore cannot bound what that asymmetry does to a transfer estimate between the two arms.** The
1665 known part of the bias runs the safe way: the removed cells are high, with a median of 563 m against the population’s lower centre, so removing them strips high negatives from the 2022 arm and makes the observed reversal smaller rather than larger.

Appendix A.6. $A(n)$. The two interventions, in full

1670 Section 4.8 states both results; the four feature-set configurations, the per-region deltas and the pooled per-target shortfalls are here.

(a) **Pooled multi-region training** (Fig. C.6). At the point estimate, training on the pooled primary populations of the other four regions never beats the best single-source transfer for any target, with shortfalls of 0.02 to
1675 0.22, and it stays 0.28 to 0.50 AUC below the within-region ceiling; for two targets it falls below the pairwise mean and below chance, though only Bejís is below chance with interval support, at 0.417 [0.369, 0.467]. Aggregation does not recover what single-source transfer loses.

(b) **Removing the direction-reversing features.** The two predictors
1680 whose signed association reverses between regions **with bootstrap support on the frames as drawn** are `elevation_mean` and `lst_anomaly_mean`.

Section 4.4 withdraws that support under an equalised frame, so this selection rule is frame-dependent and the arm below measures what the removal costs under the original protocol, not a consequence of an established reversal. The per-region signed associations and the three pair-level reversals that meet that criterion are in Appendix B, Tables B2 and B3. Retraining without them costs -0.081 of mean within-region AUC, supported in every region (per-region deltas -0.060 , -0.130 , -0.074 , -0.063 , -0.079 , every interval entirely below zero), and changes mean transfer by $+0.014$ [-0.017 , $+0.045$], which spans zero.

Two qualifications belong with those numbers rather than after them. First, the debit is not the thermal block's. Dropping elevation alone accounts for -0.061 of it, from 0.888 to 0.827. Dropping the LST anomaly alone accounts for -0.013 , from 0.888 to 0.875. Roughly three quarters of the cost is therefore the removal of a **baseline** terrain variable, not of a thermal channel. Second, both figures are post-selection. The two predictors were chosen because they reverse, using the same data on which the -0.081 and the $+0.014$ are then estimated. No correction for that selection is applied.

Configuration	Mean within-region AUC	Mean transfer AUC
full	0.888	0.541
drop <code>elevation_mean</code>	0.827	0.546
drop <code>lst_anomaly_mean</code>	0.875	0.544
drop both	0.807	0.556

Fig. C.7 shows the four configurations together. What this measures is a local cost with no compensating transfer gain. It does not measure an exchange, because the transfer side is a null on both arms. The thermal

block’s own contribution to transfer is +0.004, with an interval spanning zero (Section 4.5). Removing the reversing predictors returns +0.014, with
 1705 an interval spanning zero. Two nulls on the portability axis are not a price paid.

Appendix A.7. A(o). The thermal sign and the anomaly difference, in full

Section 4.4 states both results; the per-region values, the reciprocal stratification and the difference-instrument argument are here.

1710 **A weaker instrument does support the anomaly result.** Table B3’s note commits this paper to a difference interval on the pair as the sharper test, and it is applied here. Bootstrapping the two regions independently under the collar and differencing, four pairs have opposite-sided point estimates **and** a difference interval excluding zero, all on `1st_anomaly_mean`
 1715 and three of the four involving Evia — the largest being Bejís against Evia at $-0.191 [-0.295, -0.083]$ (Appendix A(1), Table A6). The anomaly is the one channel the frame-and-terrain mechanism of this section cannot explain, being decorrelated from elevation. Three caveats keep it weak and all three are stated rather than buried: nine of ninety feature-by-pair differences clear zero
 1720 against about 4.5 expected under the null with no multiplicity correction; the nine features are effectively two to three dimensions, as the collinearity paragraph below shows; and Evia’s own interval-support status turns on 0.003. **The honest statement is that no reversal meets this paper’s strict criterion under the collar, and that the LST anomaly differs between regions on the weaker difference instrument.** We state it that way because the alternative would be to apply a looser standard to the arm that supersedes Table B3 than to Table B3 itself. Elevation under the collar

is above 0.5 in all five regions but individually supported in only two, Muğla at 0.606 [0.525, 0.685] and Evia at 0.648 [0.550, 0.740].

1730 **The sign the five regions agree on is not the one the dryness framing predicts.** For LST the common direction is *below* 0.5 in every region, at 0.386, 0.405, 0.332, 0.286 and 0.376: a hotter pre-fire surface is associated with **less** burning, and the same holds for TVDI. It is not a lapse-rate artefact and it is not greenness acting through fuel load. On mutual
1735 adjustment the surviving channel is LST, not NDVI: holding NDVI, LST never reverses in any region, while holding LST, NDVI's own association reverses in Evia and Montiferru. Greenness is therefore not the mechanism. Two caveats bound what the sign is: in Manavgat and Montiferru it is a residual spatial gradient that disappears when distance to the nearest burned
1740 cell is stratified within the collar, and interval support is not uniform, LST being supported in four of five regions and TVDI in three, so "all five agree" is a statement about point estimates. What we can defend is that the absolute thermal channels behave here as **static land-surface descriptors** rather than as a dryness index, and that the two internally differenced channels
1745 carry no consistent cross-region direction at all. Compositing depth was not tested and remains an open alternative. Per-region stratifications are in Appendix A(k), and Section 5.2 states the consequence for the moisture-stress motivation of Section 1.2.

1750 **The same-geography arm of Section 4.10 is the most extreme case in the cohort, and it does not survive either.** A fixed study area does not mean a fixed evaluation frame: its 2022 arm has **93.2 % of cells beyond 10 km of any burned cell, median 43.6 km**, a larger far field

than any cross-region arm. Applying the same collar (`mugla_two_event_collar.csv`; the full-frame values reproduce Table B5):

	Muğla arm	share beyond 10 km	full frame	10 km collar
	2021	55.3 %	0.611 [0.529, 0.692], supported	0.606 [0.525, 0.685], supported
1755	2022	93.2 %	0.297 [0.229, 0.363], supported	0.565 [0.450, 0.677], not supported

The 2022 arm moves to the **same side of 0.5** as the 2021 arm and its interval covers chance, while the 2021 arm is nearly frame-invariant, which isolates the effect to the 2022 arm's far field rather than to the collar. **No bootstrap-supported sign reversal survives anywhere in this paper**
1760 **once evaluation frames are equalised**, between regions or between two fires in one region.

Two points bound how many independent reversals could have been counted in the first place. The channels whose reversal vanishes are the ones partly measuring terrain: current LST correlates with elevation at -0.695 to -0.125
1765 across the regions and TVDI at -0.722 to -0.298 , so their proxy strength varies fivefold and a frame that shifts the elevation distribution shifts them with it. And within the collar `fused_lst_mean` correlates with `current_lst_mean` at 0.99 to 1.00, `downscaled_lst_mean` at 0.97 to 0.99 and `current_tvdi_mean` at 0.87 to 0.98, so the four absolute channels are close to one
1770 axis and the two differenced channels correlate at 0.64 to 0.94: the "five of nine directions reverse" count of Section 4.7 counts features, not independent quantities, and **in effective dimensions it is closer to two**.

Two scope statements belong with the table. The full-frame values here are not numerically the published ones — $\rho = +0.86$ over fourteen directions
1775 against the $+0.84$ over sixteen in Section 4.6 — because the support test is itself bootstrap-dependent and at this setting three region pairs carry no jointly supported feature rather than two; the argument rests on before-and-after under identical settings, not on the published figure. And **only these diagnostics were recomputed on the collar**: the other eighteen failed
1780 on the frames as drawn and were not rerun, so their correlations against the equalised transfer vector are unknown rather than shown to be null.

The thermal block’s paired contribution to transfer, with its interval. The two matrices were differenced direction by direction. The mean is $+0.004$, but the mean is not the informative statistic here. The individual
1785 paired contributions span -0.148 to $+0.133$, with **twelve positive and eight negative** (`baseline_vs_thermal_transfer.csv`). The spread is thirty times the mean, and the sign is not a property of the block but of the pair it is asked to cross: the same six predictors that add $+0.133$ in one direction subtract 0.148 in another. A mean near zero here records cancel-
1790 lation, not consistent absence of effect. The directions are not independent, because each region appears in eight of the twenty. The interval therefore depends on what is treated as the resampling unit. All four units the design permits give the same answer:

Resampling unit	n	95 % interval on the mean paired contribution
Directions, naive	20	$[-0.027, +0.034]$
Unordered pairs, cluster bootstrap	10	$[-0.028, +0.036]$
Unordered pairs, t on pair means	10	$[-0.034, +0.042]$
Regions, leave-one-out jackknife	5	$[-0.037, +0.046]$

1795 Every interval spans zero, and the point estimate is a small fraction of each width. The per-region jackknife shows how little the mean is anchored. Holding out Manavgat, Bejís, Muğla, Evia and Montiferru in turn gives +0.0148, +0.0021, +0.0065, **-0.0081** and +0.0060. **Dropping Evia alone reverses the sign of the headline.** None of these four units propagates
1800 within-direction sampling variability. They resample between directions only.

In precision terms the transfer is worse than the ROC figures suggest. ROC-AUC is the metric used throughout this paper, for comparability with the literature, but a susceptibility surface is used as a ranked area budget, so precision-recall is the operational quantity. Across the twenty
1805 directions the thermal model’s PR-AUC averages **0.156 against a no-skill baseline of 0.136**. The mean of the twenty per-direction lifts is 1.16; the ratio of the two means just quoted is 1.146. **Six of the twenty fall below their own no-skill baseline at the point estimate, and five of those six have intervals entirely below it.** The exception is Bejís to Manavgat,
1810 at 0.034 $[0.029, 0.042]$ against a baseline of 0.038, whose interval covers the baseline and which is therefore a point-estimate case only. Only one direction, Evia to Manavgat, exceeds twice its baseline, at 0.094 against 0.038. A transferred model therefore ranks burned cells about a sixth better than

random on average, and worse than random in five directions with interval
1815 support and a sixth at the point. This is a sharper statement than the ROC
means support and it should be the one a practitioner reads.

**The within-region increment survives the same correction, and
is reported here because Section 4.2 establishes it on the frames this
section calls incomparable.** Running the baseline arm on the collar as well
1820 (`collar_increment_and_cosine.csv`, `paper/code/verify_collar_increment.py`), the thermal increment at 5 km blocking is +0.041, +0.030, +0.091,
+0.134 and +0.090 across the five regions, **positive in all five**, with a mean
of +0.077 against +0.086 on the frame as drawn. It does erode as the frame
tightens further: at a 5 km collar the mean halves to +0.041, still positive in
1825 all five. These are point estimates; the intervals in Table 1 are computed on
the frame as drawn. The paper’s one surviving predictor-level positive claim
therefore holds on the frame it argues is the correct one.

The five areas of interest are not comparable frames. Each region
is a rectangle drawn around a fire, and the rectangles differ by an order of
1830 magnitude in how much unburnt far field they enclose.

The share of modelled cells lying beyond 10 km of any burned cell is
60.1 % in Manavgat, 63.1 % in Bejís, 55.3 % in Muğla, 43.7 % in Evia and
2.1 % in Montiferru, with median distances of 13.4, 13.5, 11.3, 8.0 and 2.7
km (Appendix B, Table B6). Montiferru’s frame is fire-scale; Manavgat’s
1835 and Bejís’s are roughly three-fifths far field. The far field is not a neutral
addition. In Manavgat the median elevation of modelled cells rises from 472
m within 5 km of the fire to 955 m at 10 to 20 km and 1,273 m at 20 to 50
km, against 512 m for the burned cells themselves, in Taurus terrain that no

plausible spread model would place at risk.

1840 **Appendix B. Appendix B. All twenty transferability diagnostics**

Section 4.6 summarises this table by family. The full ranking is given here because the negative result is the point: a reader should be able to see every candidate that was tried, not only the families' best members, and to check that no diagnostic was dropped after it failed.

1845 **Table B1. All transferability diagnostics versus raw thermal transfer (20 ordered directions).** Spearman ρ with pair-based bootstrap 95 % CIs. Exp. = expected sign. Rows with $n = 12$ exist only for the four-AOI subset, because those diagnostics were never produced for Montiferru. The supported-features conditional rows use the 16 directions (8 pairs) with
1850 at least one CI-supported feature.

Diagnostic	Family	Exp.	n	dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	P(y x) conditional	+	16		+0.84 [+0.58, +0.88]	yes
Cosine, supported features	P(y x) conditional	+	16		+0.81 [+0.33, +0.88]	yes
Cosine, all 9 features	P(y x) conditional	+	20		+0.50 [−0.17, +0.83]	no
Vector	P(y x)	+	20		+0.27 [−0.36, +0.77]	no
Spearman, all 9	conditional					
Agreement count, all 9	P(y x) conditional	+	20		+0.18 [−0.40, +0.72]	no
Schoener’s D, 1-D mean	P(x y=1) niche	+	20		+0.24 [−0.45, +0.74]	no
Warren’s I, 1-D mean	P(x y=1) niche	+	20		+0.22 [−0.42, +0.73]	no
Schoener’s D, PCA-2D	P(x y=1) niche	+	20		+0.10 [−0.51, +0.68]	no
Warren’s I, PCA-2D	P(x y=1) niche	+	20		−0.07 [−0.66, +0.49]	no
Mahalanobis, burned centroids	P(x y=1) niche	−	20		−0.23 [−0.75, +0.44]	no
Domain-classifier AUC	P(x) marginal	−	20		−0.32 [−0.78, +0.33]	no
Predictor-space mean dissimilarity	P(x) marginal	−	12		−0.10 [−0.54, +0.43]	no
Predictor-space p95 dissimilarity	P(x) marginal	−	90	12	−0.08 [−0.59, +0.49]	no
Fraction inside weighted AoA	P(x) marginal	+	12		+0.22 [−0.48, +0.59]	no
Fraction inside unweighted	P(x) marginal	+	12		+0.08 [−0.89, +0.63]	no

Table note. A null row means the diagnostic was **not shown to order transfer** on this design, not that it was shown incapable of ordering it. With ten effective region pairs the power is low, and the intervals are wide
1855 enough to admit moderate true correlations in either direction. The last row is retained rather than deleted because it was computed: on two directions the statistic has no meaningful value, and reporting that is more honest than dropping the variant.

Reading the two rows that clear zero. Both are supported-feature
1860 variants, where the predictor subset is chosen by whether two regions' bootstrap intervals happen to be disjoint. That is a data-dependent selection made on the same data, uncorrected. Their all-nine-feature counterparts are in the table and both span zero. The result lives in the selection step, and Section 4.6 says so.

Equal-sample check. The families sit on unequal samples: marginal,
1865 applicability, climatic and geographic rows on twelve directions, the supported-conditional rows on sixteen, the rest on twenty. Recomputing every row on the common twelve directions reproduces the published values to 4.8×10^{-5} and leaves the ordering unchanged — the conditional rows still lead at +0.87
1870 [+0.65, +0.88] and +0.85 [+0.43, +0.88], every marginal row still spans zero. Source: `paper/diagnostics_common_subset.md`.

Appendix B.1. B2. The signed univariate associations the reversal claim rests on

Section 4.8b drops two predictors because their signed association with
1875 burning reverses between regions with bootstrap support on the frames as drawn. Section 4.4 shows that both supported elevation reversals disappear

once the frames are equalised, so this table is the evidence for the feature-removal arm and for the narrowed mechanism claim inside Contribution 1, not for a general reversal mechanism. It was computed for the frozen analysis and is reproduced here because the claim is otherwise asserted rather than shown.

Table B2. Signed univariate AUC of each predictor against burned, by region. Primary natural-vegetation population; 10-cell (~ 5 km) spatial-block bootstrap, 1000 replicates, seed 42. The AUC is never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 means lower values rank burned and is a direction rather than weakness. **Bold** marks a region whose own interval excludes 0.5.

Feature	Manavgat	Bejís	Muğla	Evia	Montiferru
elevation_mean	0.374 [0.289, 0.471]	0.643 [0.558, 0.729]	0.611 [0.532, 0.690]	0.541 [0.448, 0.626]	0.584 [0.395, 0.762]
slope_mean	0.531 [0.423, 0.642]	0.521 [0.439, 0.605]	0.637 [0.582, 0.686]	0.487 [0.418, 0.554]	0.652 [0.506, 0.771]
ndvi_mean	0.636 [0.587, 0.676]	0.559 [0.497, 0.619]	0.662 [0.616, 0.704]	0.639 [0.575, 0.701]	0.586 [0.450, 0.704]
lst_anomaly_mean	0.482 [0.428, 0.530]	0.418 [0.364, 0.480]	0.485 [0.395, 0.566]	0.640 [0.567, 0.710]	0.395 [0.285, 0.535]
current_lst_mean	0.538 [0.452, 0.621]	0.477 [0.401, 0.547]	0.325 [0.271, 0.382]	0.377 [0.301, 0.456]	0.370 [0.248, 0.513]
current_tvdi_mean	0.552 [0.460, 0.641]	0.517 [0.429, 0.595]	0.336 [0.275, 0.398]	0.362 [0.285, 0.442]	0.356 [0.233, 0.499]
tvdi_difference_mean	0.449 [0.391, 0.505]	0.512 [0.443, 0.583]	0.490 [0.396, 0.575]	0.519 [0.444, 0.589]	0.378 [0.282, 0.497]
downscaled_lst_mean	0.552 [0.466, 0.637]	0.484 [0.400, 0.560]	0.307 [0.253, 0.366]	0.377 [0.297, 0.459]	0.365 [0.240, 0.511]
fused_lst_mean	0.540 [0.454, 0.622]	0.481 [0.404, 0.551]	0.325 [0.272, 0.383]	0.376 [0.300, 0.456]	0.370 [0.248, 0.513]

What counts as a reversal. A pair of regions is called a reversal

1890 only when their signed associations point to opposite sides of 0.5 **and each region’s own interval excludes 0.5**. That is stricter than requiring the two regions’ intervals to be disjoint, and the difference matters: for `current_1st_mean` between Manavgat and Muğla the two intervals are disjoint, but Manavgat’s own interval, [0.452, 0.621], includes 0.5, so Manavgat has no
1895 established direction to reverse from. That pair is a point reversal, not a supported one.

Table B3. The cross-region reversals that meet the stricter criterion. Difference intervals are from the same paired bootstrap.

Feature	Region A	AUC	Region B	AUC	Difference [95 % CI]
<code>elevation_mean</code>	Manavgat	0.374	Bejís	0.643	+0.269 [+0.138, +0.397]
<code>elevation_mean</code>	Manavgat	0.374	Muğla	0.611	+0.235 [+0.102, +0.360]
<code>1st_anomaly_mean</code>	Bejís	0.418	Evia	0.640	+0.221 [+0.123, +0.313]

1900 Three pair-level reversals across **two** features, which is why Section 3.12 removes exactly those two. Twenty-nine further pairs reverse at the point estimate only, spread across eight of the nine features, and they are not counted. The conservative criterion costs the paper findings rather than manufacturing them: a difference interval on the pair, which is the instrument
1905 Section 4.10 uses, would support more reversals than the three listed here.

Appendix B.2. B4. The transfer matrix in precision-recall terms

ROC-AUC is reported throughout the main text for comparability with the susceptibility literature. A susceptibility surface is used as a ranked area budget, so precision-recall is the operational quantity, and at target
1910 prevalences of 3.8 to 28.7 % the two can differ sharply. Read from the same frozen step9b exports as Table B9.

Table B4. Thermal transfer, PR-AUC against the no-skill baseline. The baseline is the target's burned prevalence. Lift is PR-AUC divided by that baseline; a lift of 1 is no better than random ranking. Ordered by
1915 lift.

Direction	ROC-AUC	PR-AUC	No-skill	Lift
Evia $\rightarrow$ Manavgat	0.686	0.094	0.038	2.45
Bejís $\rightarrow$ Montiferru	0.594	0.289	0.212	1.36
Evia $\rightarrow$ Montiferru	0.647	0.283	0.212	1.34
Bejís $\rightarrow$ Muğla	0.619	0.093	0.070	1.33
Muğla $\rightarrow$ Evia	0.653	0.379	0.287	1.32
Montiferru $\rightarrow$ Bejís	0.548	0.093	0.072	1.28
Montiferru $\rightarrow$ Muğla	0.619	0.089	0.070	1.27
Evia $\rightarrow$ Muğla	0.577	0.086	0.070	1.23
Muğla $\rightarrow$ Bejís	0.583	0.088	0.072	1.22
Manavgat $\rightarrow$ Evia	0.613	0.343	0.287	1.20
Manavgat $\rightarrow$ Montiferru	0.533	0.243	0.212	1.15
Montiferru $\rightarrow$ Manavgat	0.567	0.043	0.038	1.11
Montiferru $\rightarrow$ Evia	0.586	0.317	0.287	1.11
Muğla $\rightarrow$ Montiferru	0.531	0.214	0.212	1.01
Bejís $\rightarrow$ Manavgat	0.444	0.034	0.038	0.90
Manavgat $\rightarrow$ Muğla	0.470	0.063	0.070	0.90
Evia $\rightarrow$ Bejís	0.448	0.059	0.072	0.82
Bejís $\rightarrow$ Evia	0.383	0.226	0.287	0.79
Muğla $\rightarrow$ Manavgat	0.401	0.029	0.038	0.75
Manavgat $\rightarrow$ Bejís	0.326	0.049	0.072	0.67
Mean	0.541	0.156	0.136	1.16

Six directions fall below their own no-skill baseline, and only one exceeds twice it. The six are the same six that are below chance on ROC-AUC, which is what a reversed ranking predicts in either metric. Section 4.4 shows that this count is largely a property of the evaluation frames rather than of

1920

a reversed predictor-burning relationship.

Appendix B.3. B5. The same-geography event pair, in full

Section 4.10 reports this arm and Section 4.4 withdraws its reversal as a frame artefact. The per-feature values are kept here because the arm is what motivated the frame test, and because its structural asymmetries have no analogue in the twenty-direction matrix.

Table B5. Signed univariate feature-burned AUC, Muğla 2021 versus 2022. Raw AUC against burned, never folded to $\max(\text{AUC}, 1 - \text{AUC})$; 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42 (Section 3.15). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). **Positive-carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm.** Table 1’s note sets sixteen as the floor this design supports at that blocking, so the 2022 intervals here fall below the paper’s own standard and are read as indicative, exactly as the 20-cell row of Table 1 is. The 2022 arm is additionally a single compact scar, so its eleven blocks are contiguous. Read from `paper/step9g_raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_reversal_table.csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	0.611 [0.532, 0.690]	0.296 [0.230, 0.355]	bootstrap- supported
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

1940 *Appendix B.4. B6. The evaluation frames, and the signed associations they
produce*

Section 4.4 states these results; the per-region values are here.

Table B6. Evaluation-frame geometry of the five study regions.

Primary natural-vegetation population. Distance is Euclidean to the nearest
1945 burned cell on the 500 m grid, at 0.45 km per cell. Source `aoi_frame_auc.`
`csv`; recomputable by `paper/code/verify_aoi_frame.py`.

Region	cells	burned	median distance to burned	share beyond 10 km
Manavgat	20,511	784	13.4 km	60.1 %
Bejís	15,190	1,100	13.5 km	63.1 %
Muğla	41,730	2,911	11.3 km	55.3 %
Evia	9,298	2,664	8.0 km	43.7 %
Montiferru	2,544	539	2.7 km	2.1 %

Table B7. Signed univariate AUC, frame as drawn against a 10 km collar. Point estimates; the intervals that decide the reversal question are given in the text below and in `collar_frame_bootstrap.csv` (10-cell blocks, 1000 replicates, seed 42). Signed and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is a direction, not weakness. The collar drops no burned cells in any region.

Signed univariate AUC	Manavgat	Bejís	Muğla	Evia	Montiferru	straddles 0.5
elevation, full frame (Table B2)	0.374	0.643	0.611	0.541	0.584	yes
elevation, 10 km collar	0.561	0.614	0.606	0.648	0.581	no
current LST, full frame	0.538	0.477	0.325	0.377	0.370	yes
current LST, 10 km collar	0.386	0.405	0.332	0.286	0.376	no
current TVDI, full frame	0.552	0.517	0.336	0.362	0.356	yes
current TVDI, 10 km collar	0.392	0.454	0.342	0.250	0.361	no

Table B8. Region summary: populations and gate outcomes. Counts from each region’s Step 8A dataset statistics; gate fractions from each region’s burned-landcover gate output. TSG = the primary natural-vegetation population. The TSG columns use the canonical modelled population, `burnable_tree_shrub_grass` and `valid_for_modeling == True`, which is the population every model in this paper was fitted and scored on.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned natural- veg frac- tion	Grade verdict
Manavgat 2021	24,450	24,087	796	0.033	20,511	784	0.038	0.984	pass
Bejis 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991	pass
Mugla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958	pass
North Evia 2021 (ex- tended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945	pass
Monte Lezoni 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723	pass

Table B9. Cross-region transfer matrix, thermal model, TSG population. Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after region-wise z-scoring ($\lambda = 10^{-5}$).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.326 [0.305, 0.349]	0.477 [0.451, 0.502]	0.511 [0.484, 0.534]
Bejís→Manavgat	0.444 [0.408, 0.480]	0.457 [0.420, 0.497]	0.555 [0.528, 0.583]
Manavgat→Muğla	0.470 [0.451, 0.488]	0.431 [0.411, 0.449]	0.443 [0.423, 0.461]
Muğla→Manavgat	0.401 [0.378, 0.426]	0.559 [0.531, 0.587]	0.560 [0.535, 0.587]
Manavgat→Evia	0.613 [0.593, 0.631]	0.542 [0.520, 0.565]	0.539 [0.518, 0.561]
Evia→Manavgat	0.686 [0.653, 0.716]	0.516 [0.489, 0.544]	0.527 [0.500, 0.553]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.512, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.549 [0.524, 0.575]	0.549 [0.525, 0.573]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.567 [0.539, 0.594]	0.573 [0.538, 0.609]	0.606 [0.574, 0.639]
Manavgat→Montiferru	0.533 [0.488, 0.580]	0.586 [0.540, 0.629]	0.592 [0.550, 0.631]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.581 [0.539, 0.623]

Table B10. The most and least environmentally similar pairs, on both frames. Schoener’s D is computed over burned cells only and is therefore collar-invariant. Transfer values are the two ordered directions of each pair; ranks are out of the twenty directions on the equalised frame. As-drawn transfer is read from Table B9, collar transfer from `aoi_frame_transfer.csv`.

	Manavgat–Muğla	Bejís–Montiferru
Schoener’s D , mean 1-D	0.826 (highest)	0.479 (lowest)
per-feature D	0.77 to 0.89	0.23 to 0.77
transfer, frames as drawn	0.470, 0.401	0.594, 0.548
transfer, 10 km collar	0.551, 0.510	0.669, 0.624
rank of 20 on the collar	5th, 2nd from bottom	15th, 11th
target cells inside the AoA	0.875, 0.531	—

Appendix C. Appendix C. Protocol detail

1975 These protocols are given here in full rather than in Methods, because each is a specification a reader needs only when checking the corresponding result, and none is needed to follow the argument. Sections 3.2, 3.3, 3.11 and 3.15 state in summary what each does.

1980 *Appendix C.1. C.1 Burned-area label, the reconstructed analysis grid, and the admissibility gate*

Labels come from MODIS MCD64A1 Collection 6.1 (MODIS/061/MCD64A1) retrieved through Google Earth Engine [11], whose omission and commission characteristics [4] bound every model here. The product is exported onto the 30 m EPSG:4326 reference grid used by the rest of the pipeline, which
1985 duplicates each native ~ 500 m observation across the 30 m pixels beneath it.

The analysis grid is therefore **reconstructed rather than native**: the 30 m grid is partitioned into non-overlapping blocks of $\text{round}(500 / 30) = 17 \times 17$ pixels, giving a nominal cell edge of 510 m, and each cell is identified
1990 by its integer block indices `row_500m` and `col_500m`. This approximates the

MODIS sinusoidal cell rather than reproducing it, being anchored to the Landsat/EPSG:4326 reference grid, and marginal blocks are truncated. The cell is square in degrees but not on the ground, about 510 m north to south and 390 to 407 m east to west, so its area is 0.199 to 0.208 km² against a MODIS cell's 0.250 km²; every block-size label in this paper is therefore the north-south dimension. The companion paper develops what follows from that.

A cell's representative burn date is the **mode** of the positive sub-pixel day-of-year values in the block, tested against the region's label window. Because the exported raster carries no out-of-window positives, in this dataset a single in-window positive sub-pixel makes the cell burned. The fraction of positive sub-pixels agreeing with the modal date is recorded as **burn_date_pixel_agreement_fraction** but no agreement threshold is imposed. The label never affects a cell's eligibility for modelling: unburned, all-no-data and out-of-window cells all remain as the negative class.

Two safeguards are recorded rather than assumed. Cells that burned inside a region's own predictor window are removed from its analysis universe rather than counted as unburned, which ran for Muğla, Evia and Montiferru, is recorded as not run for Manavgat and has no recorded status for Bejís. Burning in earlier years is screened for no region in the five-region cohort; the only historical exclusion in the study removes the 2021 Muğla scar from the 2022 event-relative experiment of Section 3.15.

Before any modelling each region passes a gate that answers one question: of the cells labelled burned, what fraction is dominated by natural vegetation? A region is admitted as a wildfire candidate when that fraction

reaches 0.50 and at least 30 burned cells are present, and is rejected as a cropland-dominated control when the cropland fraction reaches 0.50 instead. The gate uses ESA WorldCover classes aggregated to the same reconstructed cells. Its purpose is to separate burned area produced by natural-fuel combustion from burned area produced by post-harvest stubble burning, which MCD64A1 does not distinguish. Verdicts are reported in Section 4.1.

Appendix C.2. C.2 Transferability diagnostics versus transfer

Twenty candidate diagnostics from five families are each rank-correlated against the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap. The families are marginal predictor-distribution measures $P(x)$, burned-niche overlap $P(x|y=1)$, fire-regime label-pattern structure $P(y)$, and conditional feature-response direction $P(y|x)$.

The **marginal** family includes area-of-applicability-style dissimilarity in scaled, importance-weighted predictor space [23, 24, 18], climatic and geographic distance, and a learned domain classifier trained to separate source from target cells. The **niche-overlap** family includes Schoener’s D [36] and Warren’s I [44] over the burned cells of each region. The **regime** family includes burned-patch component counts and effective component counts. The **conditional** family is the sign-agreement index: the fraction of predictors whose signed univariate association points the same way in both regions, computed over all predictors and over the subset whose associations are individually interval-supported.

Two properties of this design are stated in advance because they bound what it can show. The candidate set was fixed in a project analysis log before

the correlations were computed; that log is not a formal pre-registration and no independent timestamped registration exists. And with five regions the effective sample is ten unordered pairs, so nineteen computed variants are evaluated at that power with no family-wise error control claimed; where a
2045 diagnostic clears an interval test but not a Bonferroni threshold, both are reported.

Appendix C.3. C.3 Same-geography event-to-event comparison (Muğla 2021 versus 2022)

Muğla is the one region for which a second fire event is analysed on the
2050 identical AOI and analysis grid, which allows the direction of feature-label associations to be compared with geography held fixed. The 2022 experiment is registered with windows anchored to its own event: a 58-day predictor window closing the day before ignition and a 49-day label window opening on it, matching the 2021 durations exactly.

**This is not a clean temporal-transfer design and is not presented
as one.** The 2022 event ignites about six weeks earlier in the season than the 2021 event, so calendar year and seasonal phase are confounded and no difference can be attributed to the year alone. The registered claim is same-geography event-to-event. The pair is deliberately kept out of the twenty-
2060 direction matrix, and that exclusion is enforced in released code rather than merely asserted here.

Two further properties bound the comparison. A historical-burn exclusion applies to the 2022 arm alone, masking every cell that burned in 2021 so the previous year’s scar does not enter the following year’s analysis: it removes
2065 3,073 cells, of which 2,941 belong to the primary population, taking it from

41,730 rows to 38,790, a drop of 7.0 %. The two arms therefore share 38,789 of 38,790 cells and three byte-identical static predictors, and in the 2022-to-2021 direction every target positive lies outside the source training population. Section 4.10 states what follows for how these two numbers may be read.

2070 **Table C1. Study regions, areas of interest and temporal windows.** Bounding boxes are in EPSG:4326, as registered. Window lengths in brackets are inclusive day counts. The baseline years are the four window-symmetric years preceding each predictor window.

Region	Bounding box (lon min, lat min, lon max, lat max)	Predictor window	Label window	Baseline years
Manavgat 2021 (Türkiye)	31.05, 36.72, 31.85, 37.35	2021-06-01 to 2021-07-27 (57 d)	2021-07-28 to 2021-08-31 (35 d)	2017, 2018, 2019, 2020
Bejís 2022 (Spain)	-1.05, 39.68, -0.35, 40.15	2022-06-15 to 2022-08-14 (61 d)	2022-08-15 to 2022-09-30 (47 d)	2018, 2019, 2020, 2021
Muğla 2021 (Türkiye)	27.1, 36.6, 28.9, 37.45	2021-06-01 to 2021-07-28 (58 d)	2021-07-29 to 2021-09-15 (49 d)	2017, 2018, 2019, 2020
North Evia 2021 (Greece)	23.05, 38.55, 23.85, 39.15	2021-06-05 to 2021-08-02 (59 d)	2021-08-03 to 2021-09-30 (59 d)	2017, 2018, 2019, 2020
Montiferru 2021 (Italy)	8.45, 40.05, 8.75, 40.27	2021-05-25 to 2021-07-23 (60 d)	2021-07-24 to 2021-08-31 (39 d)	2017, 2018, 2019, 2020

2075 *Appendix C.4. Predictor variables*

Ten predictors are used, four baseline and six thermal, all summarised per cell over the predictor window, through the chain shown in Fig. C.2. All optical and thermal predictors come from Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2), quality-screened per pixel from the QA_PIXEL
2080 band with the water bit deliberately preserved; the coarse-resolution thermal input is MODIS/061/MOD11A1.

- **NDVI**, the predictor-window median of Landsat surface reflectance, is the baseline’s one time-varying member.
- **Elevation** and **slope** come from the Copernicus DEM GLO-30.
- 2085 • **Land cover** is ESA WorldCover v200 [48], entering as the dominant class code.
- **Current LST**, the predictor-window median of Landsat surface temperature in °C.
- 2090 • **LST anomaly**, a z-score of the current-window LST median against the four baseline years, set to no-data where the baseline standard deviation is below 1.0 °C or observations are too few.
- 2095 • **TVDI** and **TVDI difference**. The Temperature-Vegetation Dryness Index [35] is computed in the LST-NDVI feature space, with wet and dry edges taken as the 2nd and 98th LST percentiles within each of twenty NDVI bins and the index clamped to [0, 1]. The edges are percentiles of the LST values a given scene contains, so they are fitted per AOI and per window, and a TVDI of 0.5 denotes a different physical moisture state in each region. The companion paper reports what that scene dependence does and does not explain. **TVDI difference** 2100 **ence** is the raw anomaly of that index against the same four window-symmetric baseline years used for the LST anomaly: `tvdi_difference = current_tvdi - mean(baseline_tvdi)`, in index units rather than standard deviations, which is why it is reported alongside the z-scored channel rather than in place of it.

2105 • **Downscaled LST and fused LST.** A MODIS-to-Landsat downscal-
 ing model is trained on the predictor window and applied to the full 30
 m grid. The fused product equals observed Landsat LST wherever that
 is valid, and the downscaled surface only where it is not. Gap-filling
 therefore never replaces or blends a valid observation. The gap-filled
 2110 share is 0.11 % to 9.70 % by region. The downscaler’s own inputs
 include coordinates, which is the one route by which a coordinate-
 derived surface re-enters a feature set from which Section 3.14 excludes
 coordinates. Appendix A(g) reports the increment without these two
 channels.

2115 *Appendix C.5. C.4 Limitations, in full*

Section 5.9 states each limitation; the elaborations are here.

- (i) **No meteorological covariates** enter the models, so we cannot say
 how local skill and portability behave for a mixed thermal-plus-weather pre-
 dictor set.
- 2120 (ii) **The same-geography comparison covers one region only**, and
 even there year and seasonal phase are confounded, a confound that cannot
 be resolved in this study area for the reason given in Section 5.2. Its 331
 burned cells also leave the thermal reversals unresolved at interval level,
 and the pair holds place fixed but not population. Those two arms are
 2125 additionally the only transfer directions here computed by us rather than
 read from the pipeline author’s frozen export, with his unmodified code and
 the same pinned environment.
- (iii) **All labels derive from a single burned-area product**, MCD64A1
 [10], whose omission and commission characteristics [4] bound every model

2130 evaluated here. No second burned-area product covers 2021 and 2022 at this resolution; the companion paper reports what an independent active-fire observation says about the omission concern.

(iv) **Evia remains the most imbalance-atypical population** even after the AOI extension, at a TSG prevalence of 0.287 against 0.038 to 0.072
2135 in three of the others.

(v) **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology, and distinguishing them requires multi-year labels.

(vi) **Cross-region point estimates carry an implementation tolerance** of roughly ± 0.02 to 0.03 across scikit-learn versions. All reported
2140 numbers are fixed to one verified version, but exact reproduction elsewhere requires the archived environment.

(vii) **The diagnostic correlations rest on an effective sample of ten region pairs.** Both the successes and the failures of Section 4.6 should
2145 be read at that power.

(viii) **Manavgat’s atypical transfer behaviour remains unexplained.** It is where the conditional diagnosis bites hardest and where feature removal recovers most. Three candidates have now been tested and none survives: its meteorology, which was not extreme; the quality screening of its coarse
2150 thermal input, which propagates widely but moves no signed association by more than +0.0003 (Section 4.9, Appendix A(e)); and the evaluation frame, which explains its elevation figure but not its transfer behaviour (Section 4.4). With one fire season per region the remaining candidates are not separable in this design.

2155 (ix) **The interval-support counts are less stable than the point estimates behind them.** Several verdicts sit within a thousandth of their reference value, and at 1 km blocking the published split of ten positive, seven negative and three uncertain turns on a lower bound of -0.00045 . The point estimates and the sign pattern are stable; the counts are not. Every sentence
2160 in this paper that leans on an exact count of supported directions should be read at that precision. (x) **The five areas of interest are not comparable frames, and this cohort cannot fully repair it** (Section 4.4). We report the equalised arm alongside the frame-as-drawn arm rather than replacing one with the other, because the collar radius is itself a choice and 5 km and
2165 10 km do not agree exactly (0.608 against 0.617). The deeper limitation is that the frames were fixed upstream of this work, in **repo/**, so we can restrict them but not extend them; a region whose rectangle is already fire-scale, Montiferru, cannot be given a far field for symmetry. Any future cohort should fix the frame by an explicit accessible-area rule [3] before any predictor
2170 is computed, and we treat that as the main design lesson of this paper.

CRedit authorship contribution statement

Yunus Emre Cogurcu: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review and editing, Supervision. **Emrehan Metin:** Software, Data curation, Investigation, Val-
2175 idation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

2185 All satellite inputs are public and are obtained through Google Earth Engine. The processing pipeline, its configuration and the frozen numeric outputs are publicly available; the repository, the commit of record and the licence are given in the data and code availability statement of the Methods, together with the three components that fall outside that release. No digital object identifier is minted and no archival deposit exists.

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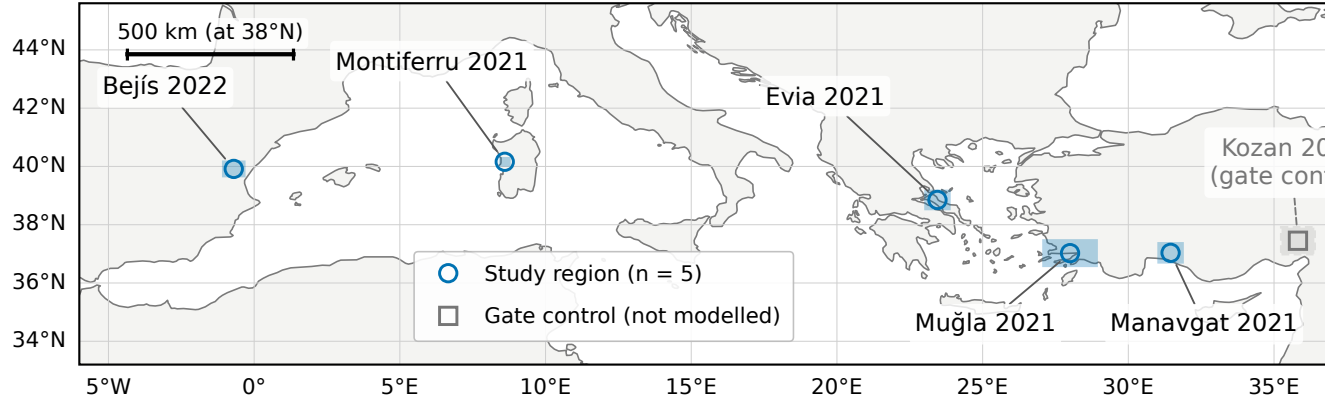


Figure C.1: **Study regions and the gate negative control.** Locations of the five wild-fire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejís 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table 1 to 5×10^{-3} degrees.

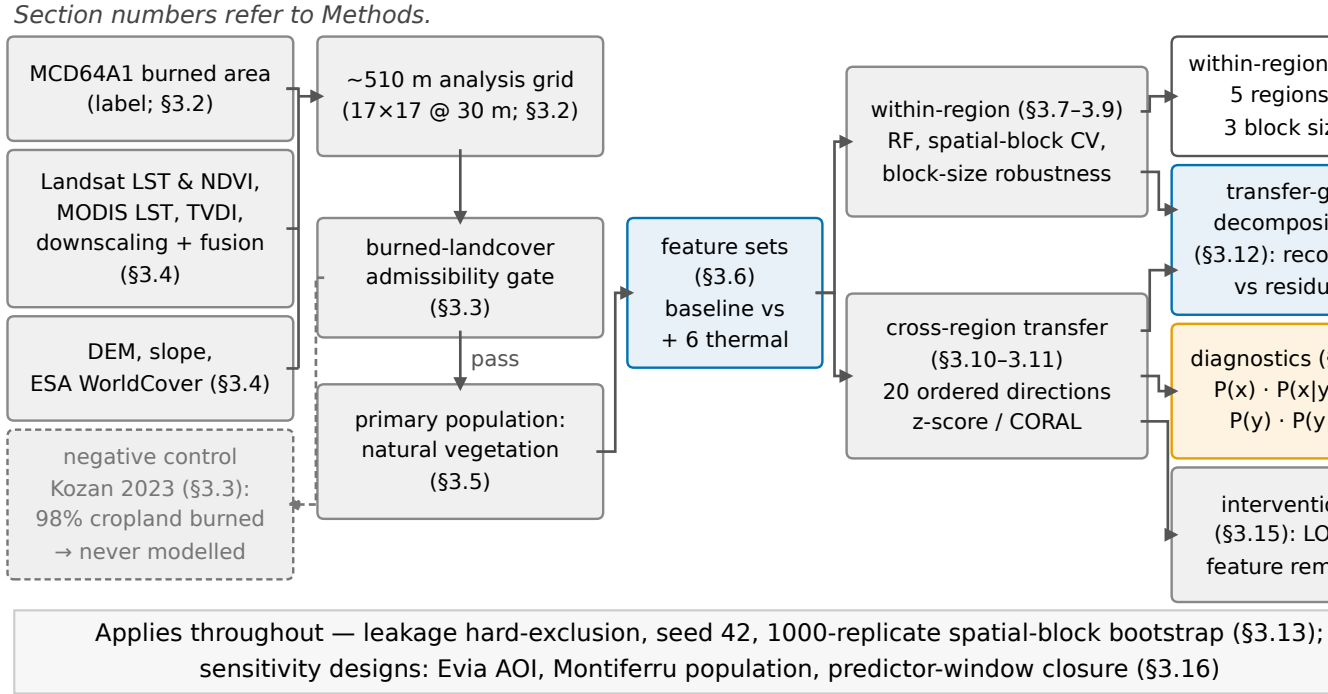


Figure C.2: Processing pipeline and evaluation programme. Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed ~ 510 m analysis grid (17×17 pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.12). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (§3.16). Section references are verified against the Methods heading list when the figure is built.

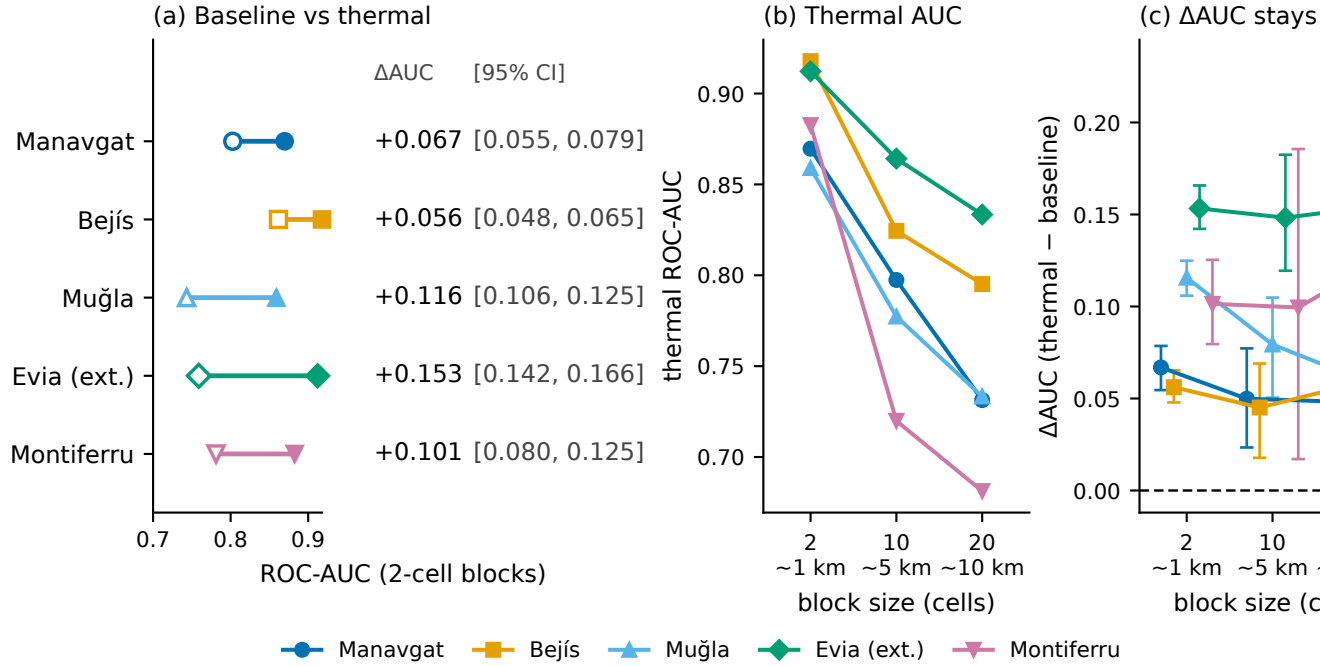


Figure C.3: **The thermal increment replicates in all five regions and survives coarser spatial blocking.** Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell (~ 1 km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired ΔAUC and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells, $\approx 1, 5$ and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment ΔAUC against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region-block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 3, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes** (2, 10, 20 cells), and the offset is smaller than half the spacing between block sizes, so the groups cannot merge. Colour and marker shape both encode region, so the panels remain readable in greyscale.

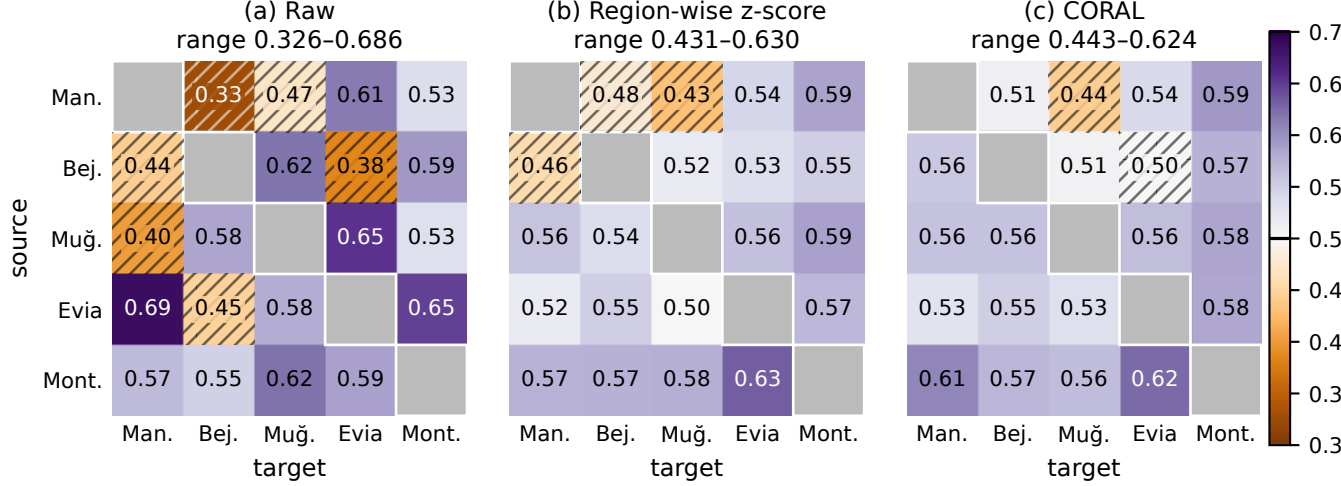


Figure C.4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source–target directions among the five regions, natural-vegetation primary population. **(a)** Raw transfer, no adaptation. **(b)** After region-wise z-score standardisation. **(c)** After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation compresses the spread from 0.326–0.686 (raw) to 0.431 to 0.630 (z-score) and 0.443 to 0.624 (CORAL). It pulls the extremes toward chance rather than lifting transfer. Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so two cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance) and z-score Evia→Muğla is 0.5013 (un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.7:1). Confidence intervals for these directions are given in Table 4.

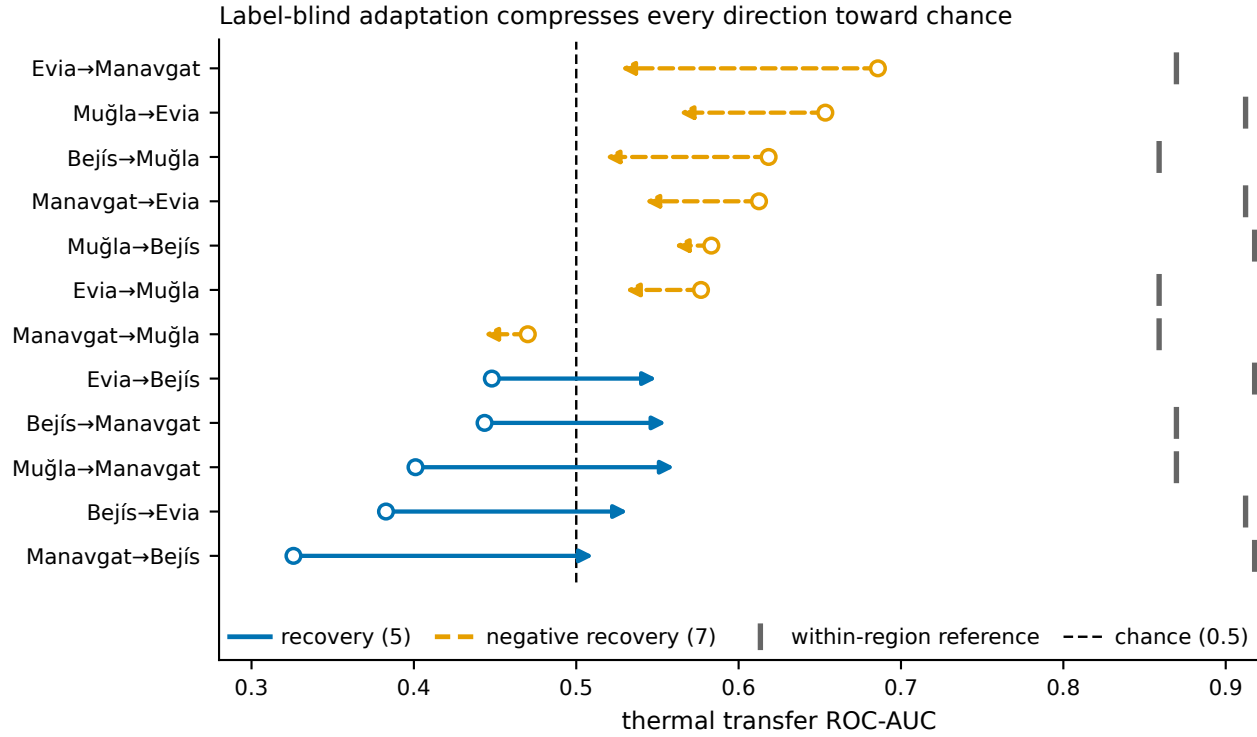


Figure C.5: **Label-blind adaptation regresses transfer onto the level a model reaches on an unseen scar, rather than recovering it.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves *away* from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. The pattern is the point: all six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.470 and is pushed further down to 0.443. The worst case is Evia→Manavgat, where raw transfer of 0.686 is dragged down to 0.527 against a within-region reference of 0.870, a recovered fraction of -0.862 . No arrow reaches its within-region tick.

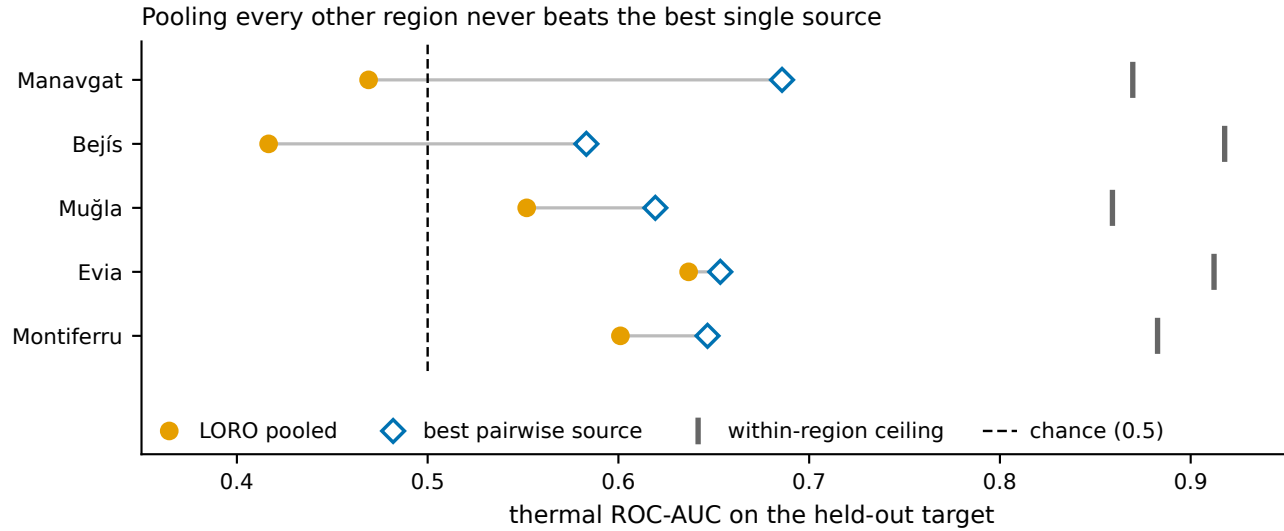


Figure C.6: **Pooling every other region never beats the best single source.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in all five targets without exception. Adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.469, Bejís 0.417), and only for Bejís does the interval lie entirely below chance, while a single well-chosen source reaches 0.686 and 0.583 respectively. Every point remains far below the within-region ceiling.

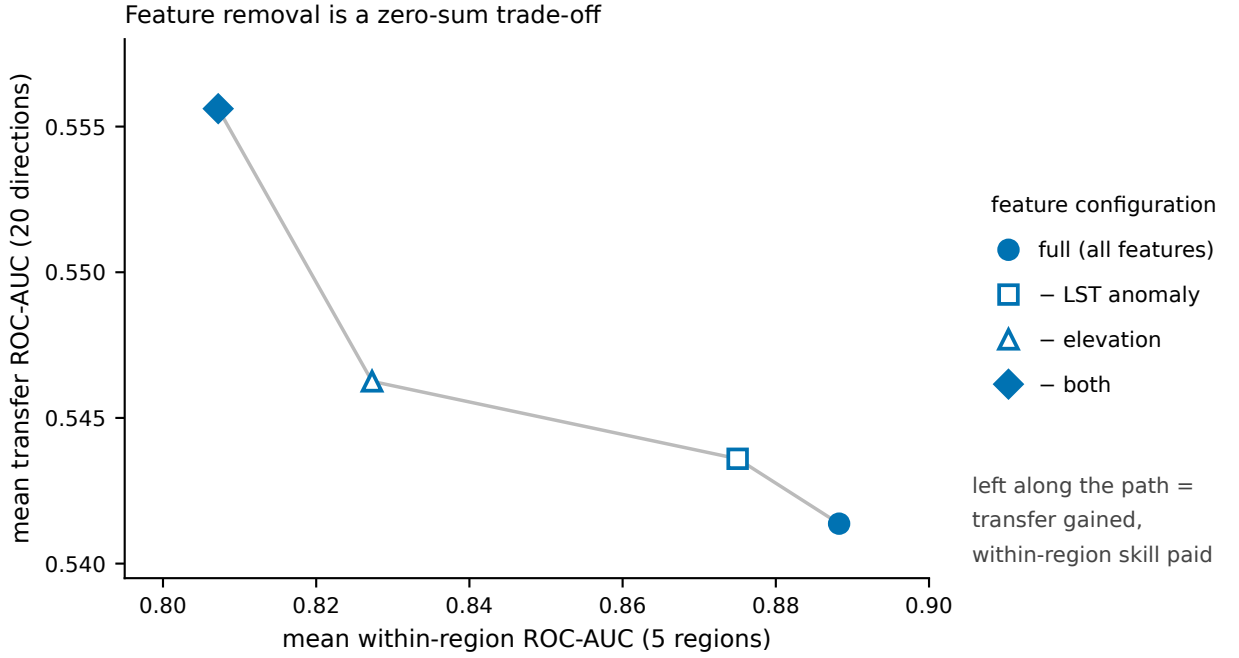


Figure C.7: **Removing the direction-reversing features costs locally and returns nothing measurable on transfer.** Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, transfer rises monotonically ($0.541 \rightarrow 0.556$) while within-region skill falls monotonically ($0.888 \rightarrow 0.807$). Both monotonicities are verified when the figure is built, so the figure cannot silently contradict the claim. The within-region cost is 0.081 AUC and is supported in every region. The change in transfer is +0.014 AUC on an interval that spans zero, so no exchange between the two is demonstrated: a local cost is measured and no compensating transfer gain is.

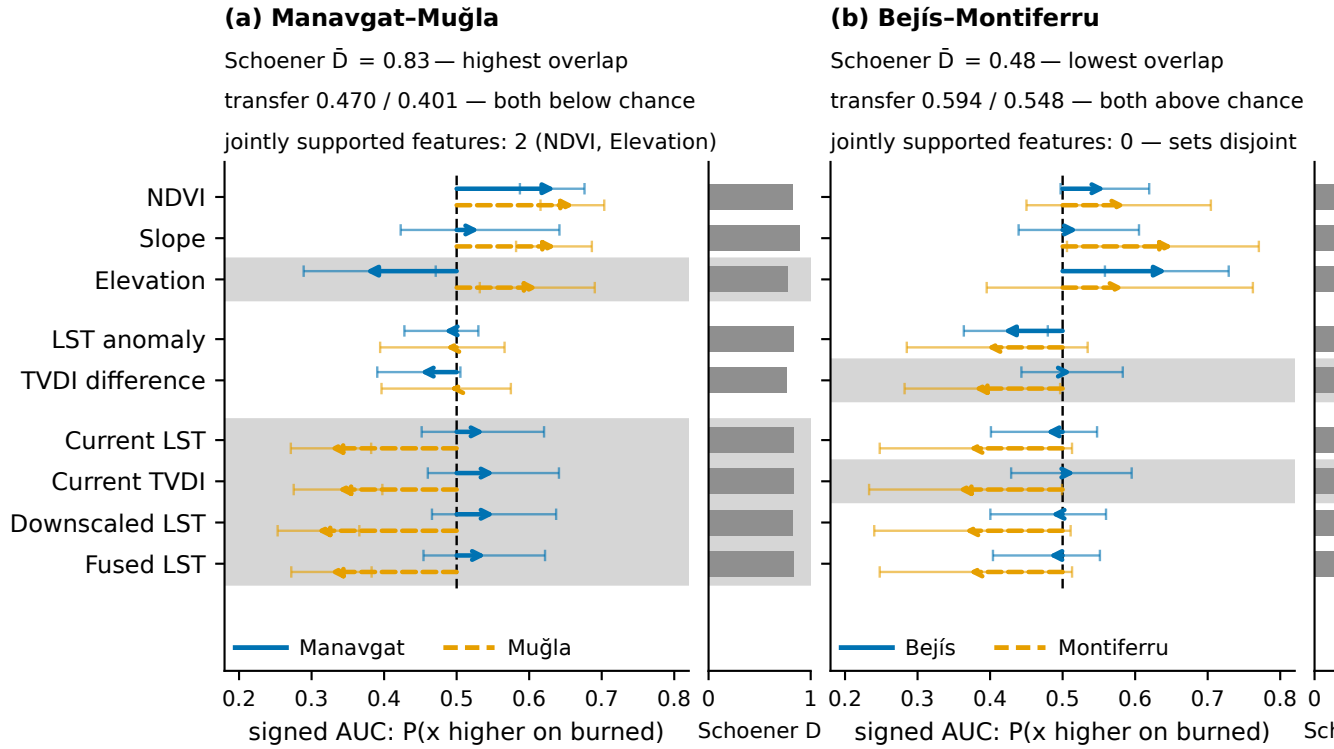


Figure C.8: **Burned-niche overlap does not determine transfer: the contrast pair.** Per-feature signed univariate AUC, the probability that a predictor takes a higher value on a burned cell than on an unburned one, for the two most informative region pairs, natural-vegetation primary population. Arrows run from chance (0.5) to the observed value, so their direction is the sign of the association and their length its strength; whiskers are 95% spatial-block (≈ 5 km) bootstrap intervals. Within each pair the two regions are distinguished by colour, by line style (solid for the first region named, dashed for the second) and by vertical offset; line style is included because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. **A filled arrowhead means that region's 95% interval excludes 0.5, so the direction is statistically supported; an open arrowhead means the interval contains 0.5 and the direction is not supported.** Rows shaded grey are features whose sign disagrees between the two regions. Grey side bars are the per-feature Schoener's D overlap. **(a)** Manavgat–Muğla, the pair with the *highest* burned-niche overlap ($\bar{D} = 0.83$), transfers below chance in both directions (0.470 and 0.401). **(b)** Bejís–Montiferru, the pair with the *lowest* overlap ($\bar{D} = 0.48$), transfers above chance in both directions (0.594 and 0.548). High niche overlap is therefore not sufficient for transfer. The arrowhead fill also exposes the coverage limit of the supported-agreement index, which is defined only on features whose direction is supported in *both* regions: there are two such features in (a) (NDVI and elevation) and **none in (b)**, where the two regions' supported sets are disjoint and Montiferru's intervals

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